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Wireless Hacking

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Scope of Wireless Hacking

- Wireless Local Area Networks
  - Components of WLANs
  - WiFi Security Standards
- Wireless Attacking Techniques and Tools
  - Rogue APs
  - Wardriving
  - DoS/ DDoS
  - Key Cracking
- CIA Attacking Tools
- Defenses
  - Dos and Don’ts
  - Best Practices





Components of a WLAN

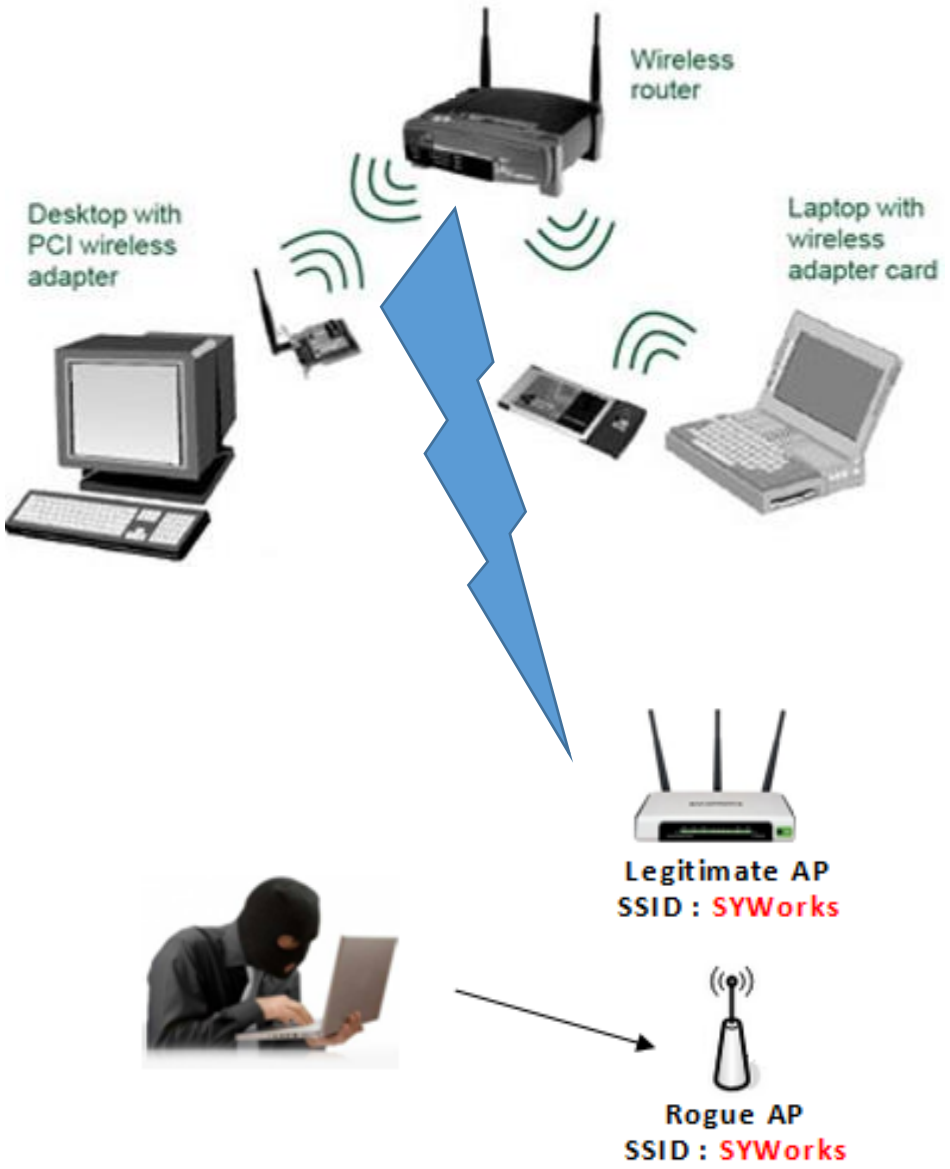
- Access Point (AP) - is essentially a hub that allows different wireless LANs (WLANs) to connect to a network backbone
- Router / Modem – modulates analog signal to encode/decode digital info; provides APs with connection to Internet
- Devices in each WLAN may be interconnected via a local wireless router



Components of a WLAN

- Wireless Adapter - transmits and receives wireless signals; configured to talk with other wireless adapters within range
- The Wireless Network Interface Card (WNIC) in your laptop is an example of a wireless adapter
- A wireless router in a WLAN can communicate with other WLANs via an AP
- APs are what attackers look for

What are SSIDs?



Service Set Identifiers

- A Service Set Identifier (SSID) is used to identify a WLAN, much the same way as a workgroup is used on a Windows network
- An SSID is configured on the AP as a unique, 1-to 32-character, case-sensitive alphanumeric name
- The AP usually broadcasts (**beacons**) the SSIDs several times a second so that users who have WNICs can see a display of all WLANs within range of the AP's signal



AP beacons SSID or SSIDs

WNIC
(No WiFi)

WNIC
(No WiFi)

WNIC
(No WiFi)

AP

One AP can beacon multiple SSIDs



Access Point

Default SSIDs

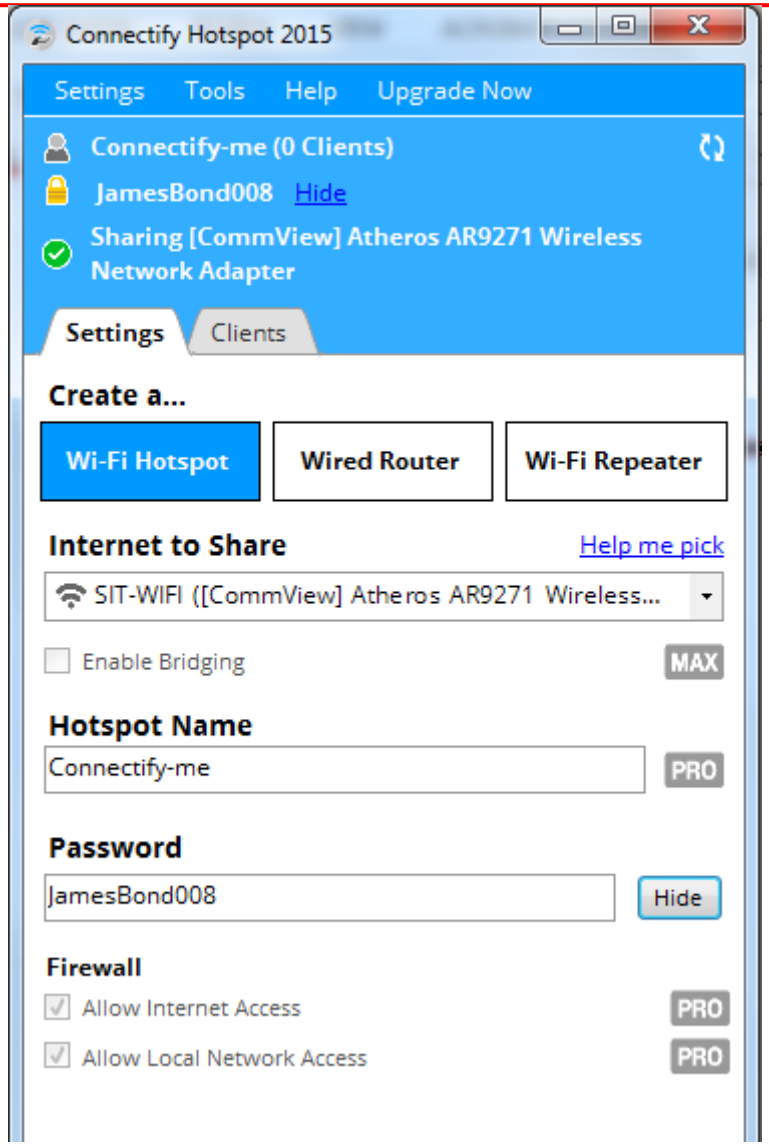
Vendor	Default SSIDs
3Com	3Com, comcomcom, 101
Apple	Airport Network
Belkin (54G)	Belkin54g
Cisco	tsunami
Compaq	COMPAQ
D-Link	WLAN, default
Dell	wireless
Intel	Intel, 101, XLAN, 195, Intel Gateway
Linksys	linksys, wireless, linksys-g
Microsoft	MSHOME
Netgear	Wireless, Netgear
SMC	WLAN, BRIDGE, SMC
Symantec	101
U.S. Robotics	WLAN, USR9106, USR5450, USR8022, USR8054

Many vendors have SSIDs set to a default value that companies never change

Rogue AP

- Unmanaged (unauthorized) AP
  - Attached to enterprise network
- A significant potential backdoor
- Very easy to install a wireless AP
  - Latest OS helps you share your internet
  - Many software helps you create a software AP
- Sometimes Malicious Intent
  - Or an unwitting impatient employee

Rogue AP Connectify





- Download Connectify Hotspot from: http://connectify-lite.en.softonic.com/

Rogue AP

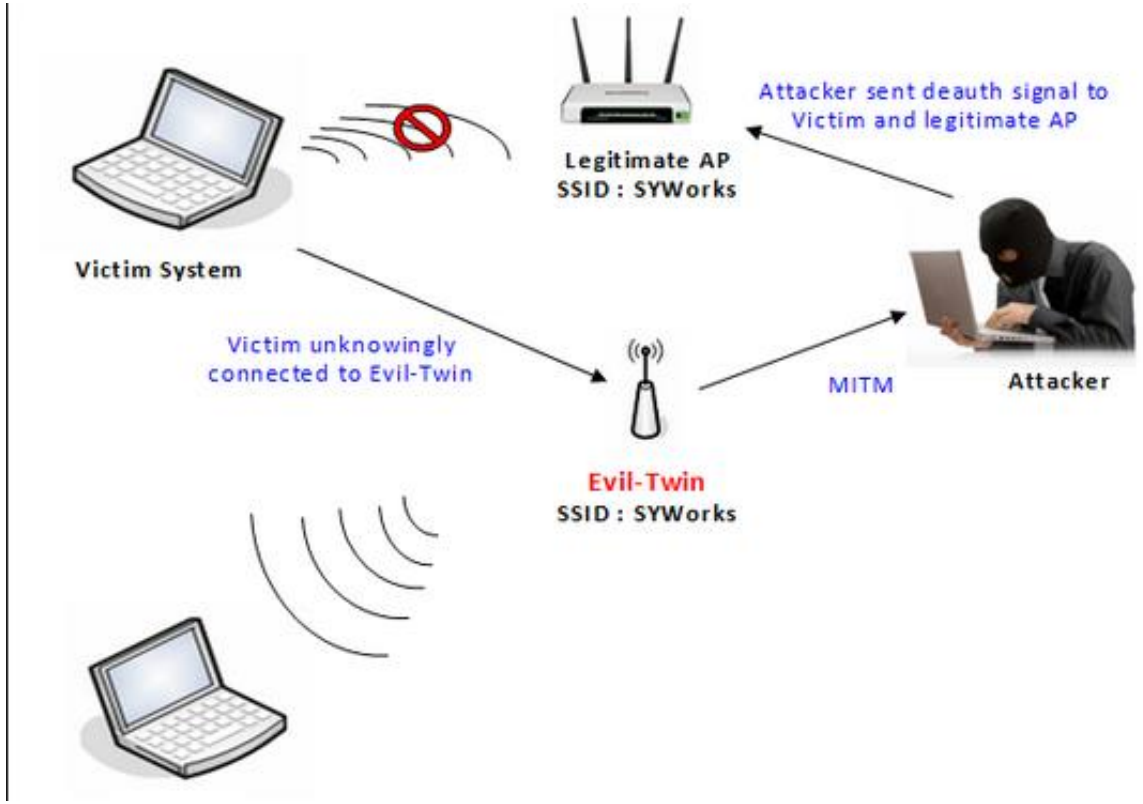
- Attacks on wired network infrastructure
  - ARP poisoning, DHCP attacks, STP attacks, DoS attacks etc
- Mapping the network for targeted attacks
- Scanning the network
- Man in the middle and data sniffing

- Some solutions
  - Network Access Control
    - Not always effective
    - MAC Address spoofing
  - Schedule periodic manual survey

Evil Twin

- Get a wireless access point
- Configure it the same way as an existing network
  - Same SSID and security settings
- Overpower the existing AP
  - Do not need to be in the same vicinity
- WiFi hotspots are easy baits

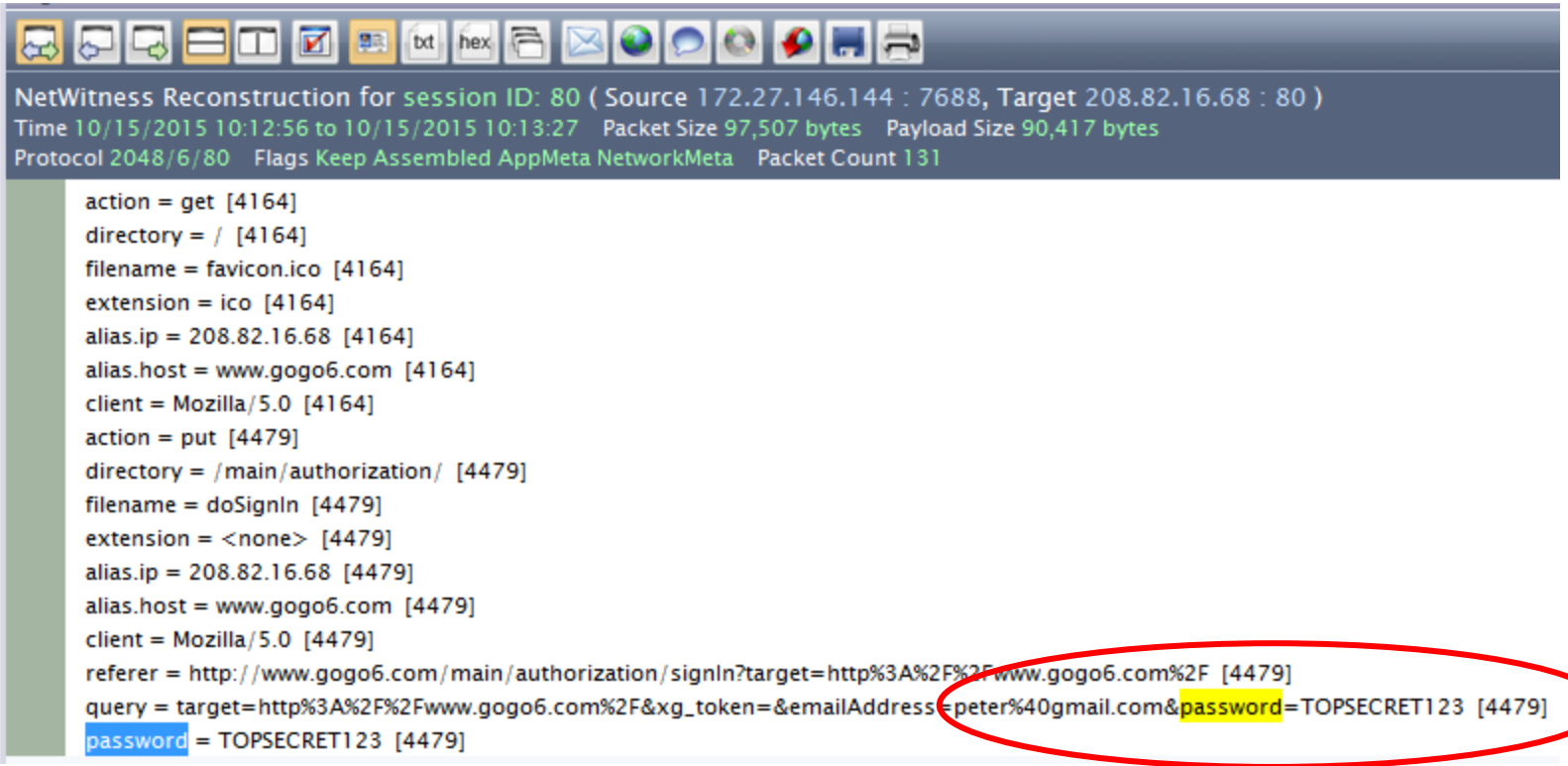
Evil Twin



- Most clients will associate to the AP with the strongest signal
- Evil Twin will then have access to the network traffic of all associated clients
- The evil twin can also use MAC Address Spoofing to trick clients into sending passwords and sensitive information

Evil Twin- MITM





Man in the Middle Attacks

- To install an evil twin, an attacker needs to know the SSID of the trusted AP and activate it without authentication
- If the AP broadcasts the SSID (or use the default one), an attacker may easily obtain the SSID by using a "sniffer"
- IEEE 802.11b standard by default stipulates one-way authentication
- That is, an AP authenticates a user, but a user does not authenticate an AP
- So a user cannot tell when an AP is not legitimate

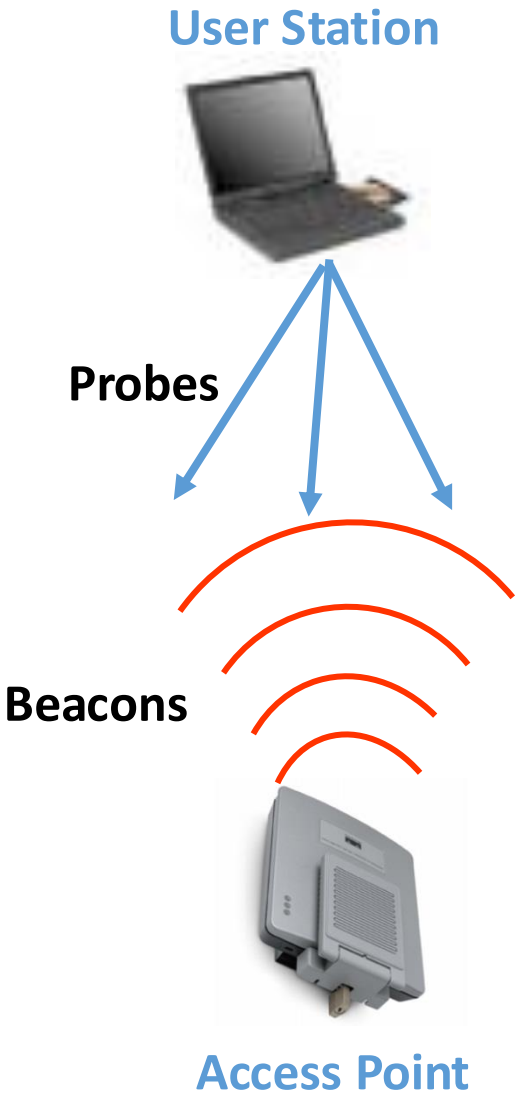
Probes and Beacons

PROBES:

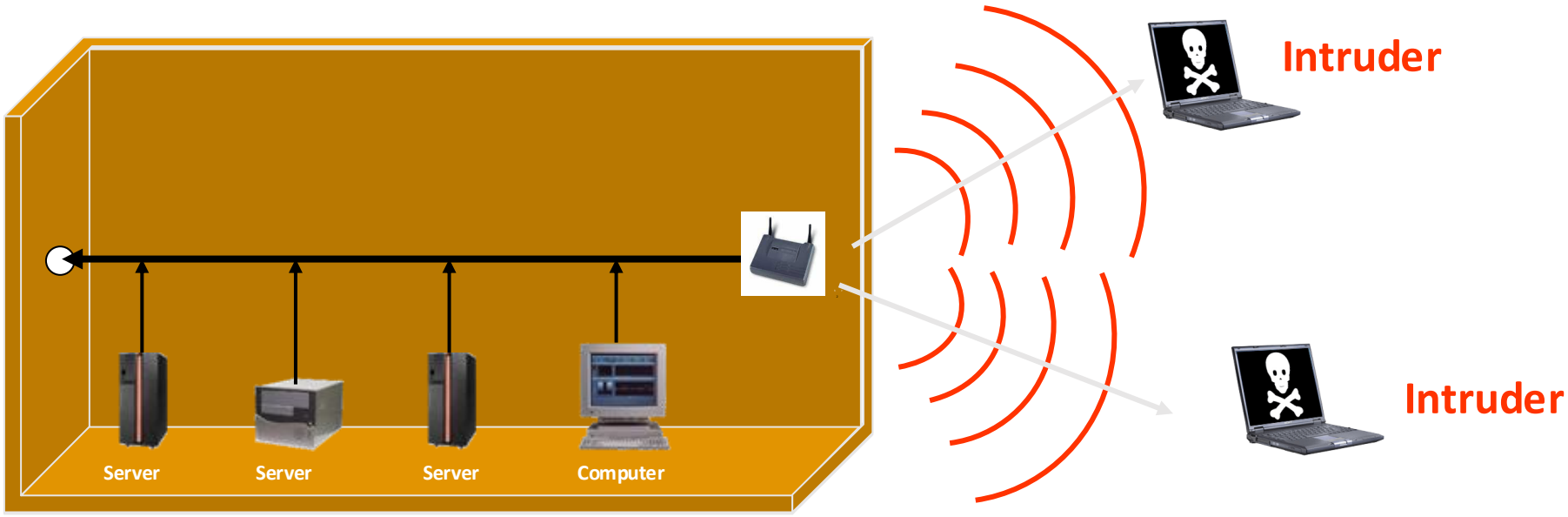
- A Station sends a probe request frame when it needs to obtain information from another station. (*For example, a station would send a probe request to determine which access points are within range.*)

BEACONS:

- The Access point (AP) periodically sends a beacon frame to announce its presence and relay information, such as timestamp, SSID, and other parameters regarding the access point

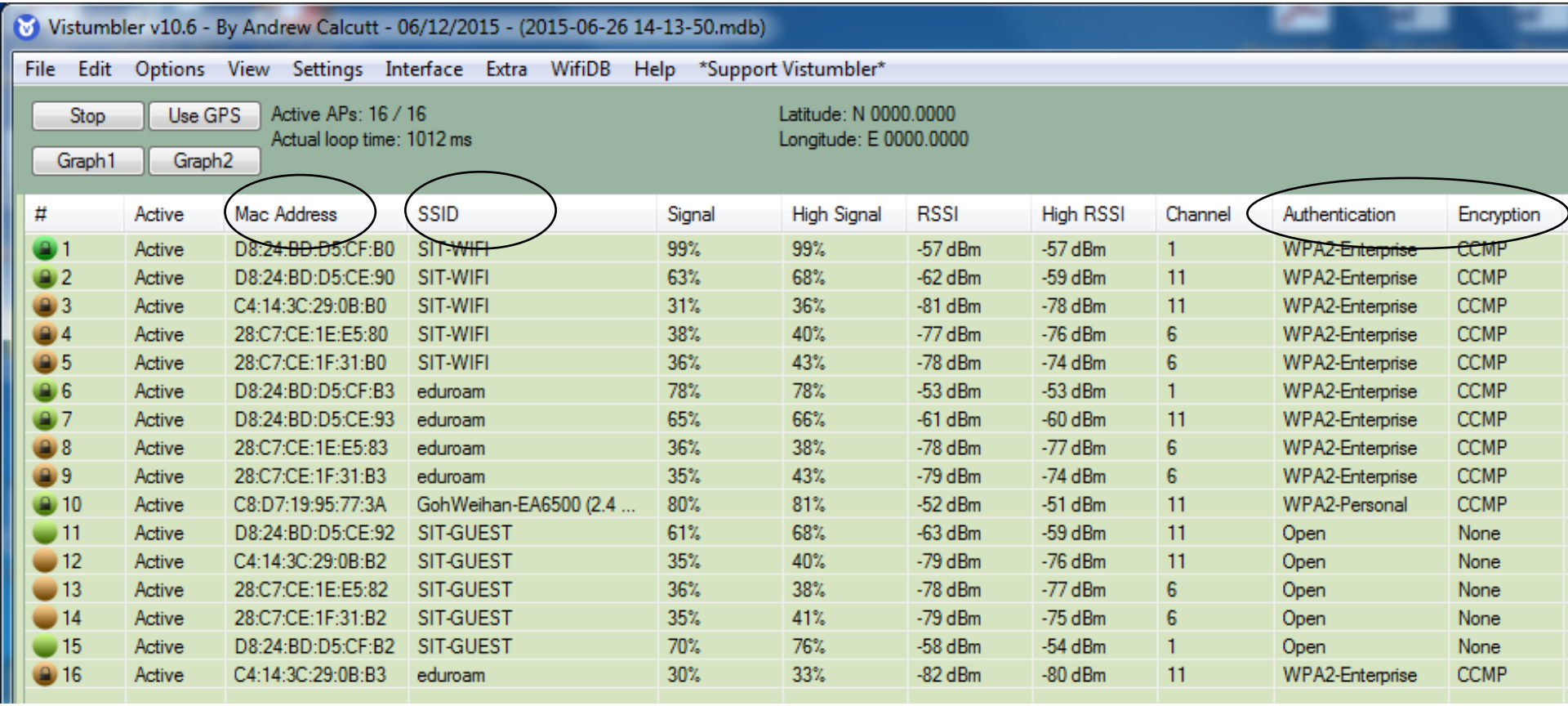


Problem: Uncontrolled RF Medium



- With AP beaconing their info, walls are no longer barriers

What can an Attacker see?



#	Active	Mac Address	SSID	Signal	High Signal	RSSI	High RSSI	Channel	Authentication	Encryption
1	Active	D8:24:BD:D5:CF:B0	SIT-WIFI	99%	99%	-57 dBm	-57 dBm	1	WPA2-Enterprise	CCMP
2	Active	D8:24:BD:D5:CE:90	SIT-WIFI	63%	68%	-62 dBm	-59 dBm	11	WPA2-Enterprise	CCMP
3	Active	C4:14:3C:29:0B:B0	SIT-WIFI	31%	36%	-81 dBm	-78 dBm	11	WPA2-Enterprise	CCMP
4	Active	28:C7:CE:1E:E5:80	SIT-WIFI	38%	40%	-77 dBm	-76 dBm	6	WPA2-Enterprise	CCMP
5	Active	28:C7:CE:1F:31:B0	SIT-WIFI	36%	43%	-78 dBm	-74 dBm	6	WPA2-Enterprise	CCMP
6	Active	D8:24:BD:D5:CF:B3	eduroam	78%	78%	-53 dBm	-53 dBm	1	WPA2-Enterprise	CCMP
7	Active	D8:24:BD:D5:CE:93	eduroam	65%	66%	-61 dBm	-60 dBm	11	WPA2-Enterprise	CCMP
8	Active	28:C7:CE:1E:E5:83	eduroam	36%	38%	-78 dBm	-77 dBm	6	WPA2-Enterprise	CCMP
9	Active	28:C7:CE:1F:31:B3	eduroam	35%	43%	-79 dBm	-74 dBm	6	WPA2-Enterprise	CCMP
10	Active	C8:D7:19:95:77:3A	GohWeihan-EA6500 (2.4 ...	80%	81%	-52 dBm	-51 dBm	11	WPA2-Personal	CCMP
11	Active	D8:24:BD:D5:CE:92	SIT-GUEST	61%	68%	-63 dBm	-59 dBm	11	Open	None
12	Active	C4:14:3C:29:0B:B2	SIT-GUEST	35%	40%	-79 dBm	-76 dBm	11	Open	None
13	Active	28:C7:CE:1E:E5:82	SIT-GUEST	36%	38%	-78 dBm	-77 dBm	6	Open	None
14	Active	28:C7:CE:1F:31:B2	SIT-GUEST	35%	41%	-79 dBm	-75 dBm	6	Open	None
15	Active	D8:24:BD:D5:CF:B2	SIT-GUEST	70%	76%	-58 dBm	-54 dBm	1	Open	None
16	Active	C4:14:3C:29:0B:B3	eduroam	30%	33%	-82 dBm	-80 dBm	11	WPA2-Enterprise	CCMP

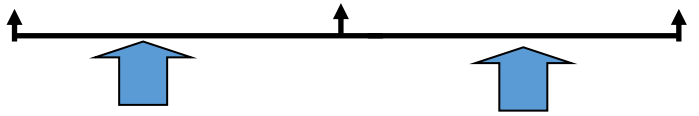
VISTUMBLER

SSID & MAC Address

- **SSID** helps find reachable APs
  - 32 byte unique Service Set Identifier of AP
  - Sent when AP receives a probe request from station
  - "Visible" in the air



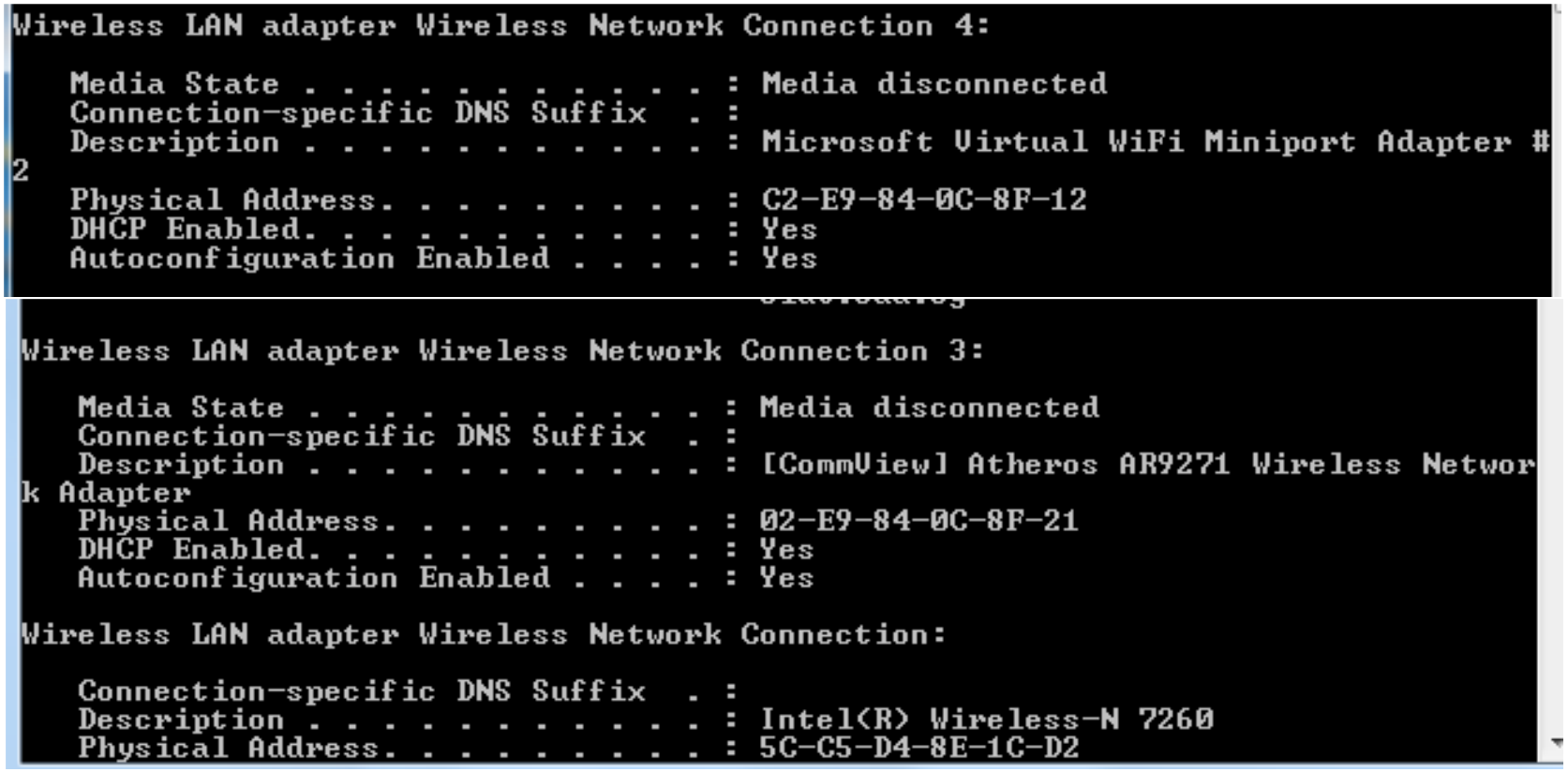
- **Media Access Control (MAC) Address**
  - To deliver traffic, a unique 48-bit identifier must be available for each network device
  - Example: 00 – 04 – 5a – 03 – 3c – 0f



(Organizationally Unique Identifier, first 3 characters)

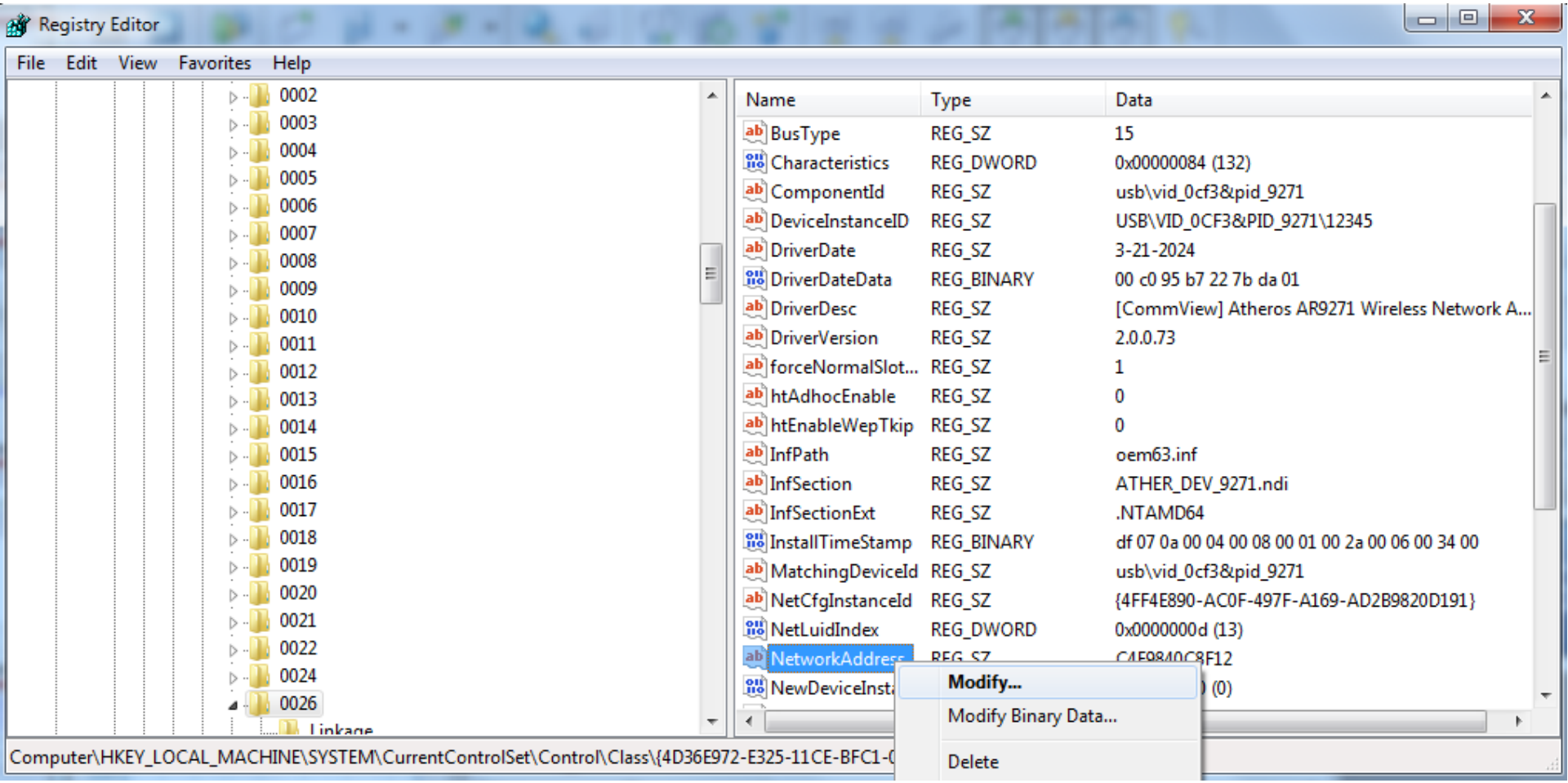
Vendor	OUI
Cisco (Aironet)	00-04-96
Agere (Orinoco)	00-02-2D
Nokia	00-e0-03
Linksys	00-04-5a

MAC Address Spoofing via Device Manager



- C> ipconfig/all
- Control Panel -> View Network Status and Task -> Change adapter settings
- First 2 chars must be ‘02’ - sets b1 bit so that MAC address is locally administered

MAC Address Spoofing via Registry



HKEY_LOCAL_MACHINE | SYSTEM | CurrentControlSet | Control | Class | {4D36E972-E325-11CE-BFC1-08002BE10318} | 0017 | NetworkAddress

MAC Address Spoofing via Tool

```
┌──(vivek㉿vivek)-[~]
└─$ sudo macchanger -s wlan0
Current MAC:   f4:f2:6d:1d:9f:47 (unknown)
Permanent MAC: f4:f2:6d:1d:9f:47 (unknown)

┌──(vivek㉿vivek)-[~]
└─$ sudo macchanger -a wlan0
Current MAC:   f4:f2:6d:1d:9f:47 (unknown)
Permanent MAC: f4:f2:6d:1d:9f:47 (unknown)
New MAC:       00:1c:95:21:98:74 (Opticomm Corpo

┌──(vivek㉿vivek)-[~]
└─$ sudo macchanger -m f4:f2:6d:ab:cd:ef wlan0
Current MAC:   00:1c:95:21:98:74 (Opticomm Corporation)
Permanent MAC: f4:f2:6d:1d:9f:47 (unknown)
New MAC:       f4:f2:6d:ab:cd:ef (unknown)

┌──(vivek㉿vivek)-[~]
└─$ sudo macchanger -p wlan0
Current MAC:   f4:f2:6d:ab:cd:ef (unknown)
Permanent MAC: f4:f2:6d:1d:9f:47 (unknown)
New MAC:       f4:f2:6d:1d:9f:47 (unknown)
```

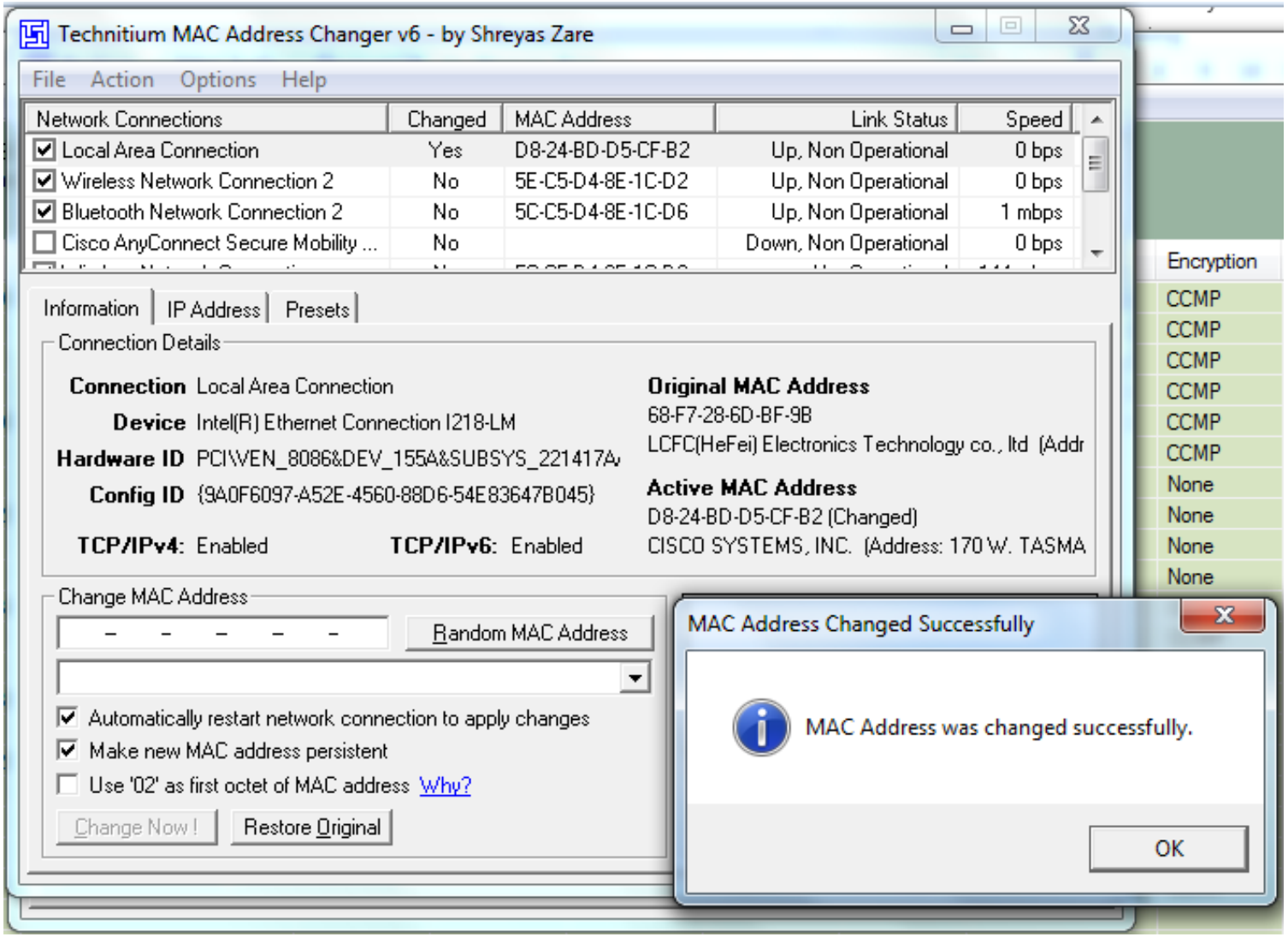
```
└─$ sudo macchanger -h
GNU MAC Changer
Usage: macchanger [options] device

  -h,  --help                  Print this help
  -V,  --version               Print version and exit
  -s,  --show                  Print the MAC address and exit
  -e,  --ending                Don't change the vendor bytes
  -a,  --another               Set random vendor MAC of the same kind
  -A                           Set random vendor MAC of any kind
  -p,  --permanent             Reset to original, permanent hardware MAC
  -r,  --random                Set fully random MAC
  -l,  --list[=keyword]        Print known vendors
  -b,  --bia                   Pretend to be a burned-in-address
  -m,  --mac=XX:XX:XX:XX:XX:XX
       --mac XX:XX:XX:XX:XX:XX  Set the MAC XX:XX:XX:XX:XX:XX

Report bugs to https://github.com/alobbs/macchanger/issues
```

MACCHANGER – KALI Linux

MAC Address Spoofing via Tool



TECHNITIUM MAC ADDRESS CHANGER

What can an Attacker Do?

- **AP Honeypot:** Configure a fake or rogue AP's MAC to divert legitimate WiFi traffic in MITM attacks, Replay attacks etc.

- **Bypass MAC Address Filtering**: Networks can use MAC address filtering, only allowing devices with specific MAC addresses to connect to a network. Spoofing an authorized device MAC can bypass this filtering to exploit free WiFi usage, for example

Access Point (AP) Honeypot

- Hacker sets up his rogue AP
  - Software is available to convert a laptop or USB WiFi adapter into a Soft Access Point (Virtual Router or Connectify)
  - The soft access point broadcasts spoofed SSID of the legitimate AP
  - Attacker may even alter the MAC to that of the legitimate AP
- User connects to the rogue AP assuming it to be legitimate
- Attacker harvests personal information from the link
- Windows 7 has a feature “SoftAP” -- which can be used for Internet Connection Sharing (use Connectify)
- netsh wlan set hostednetwork mode=allow ssid="ssid" key="passphrase" keyUsage=persistent
- Soft AP configuration tools e.g. Virtual WiFi Router, My Public WiFi, My WiFi Router

MAC Address Filtering

- MAC address filtering refers to the process of allowing or denying access to a wireless network based on the hardware address of the device attempting to connect
- Only clients with permitted MAC addresses can access the wireless network
  - List of allowable MAC addresses can be configured on the AP, or
  - it can also be configured on a RADIUS server that the AP communicates with
- Spoofing occurs when an attacker changes a device's MAC address to impersonate an authorized device on the wireless network
- Relatively easy to discover authorized addresses
  - MAC addresses are sent in the clear on wireless networks and can be sniffed

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WiFi Security Standards

WiFi Security Standards - Introduction

	WEP	WPA	WPA2	WPA3
Release Year	1999	2003	2004	2018
Encryption Method	Rivest Clipher 4 (RC4)	Temporal Key Integrity Protocol(TKIP) with RC4	CCMP and Advanced Encryption Standard	Advanced Encryption Standard(AES)
Session Key Size	40-bit	128-bit	128-bit	128-bit(WPA3-Personal) 192-bit(WPA3-Enterprise)
Clipher Type	Stream	Stream	Block	Block
Data Integrity	CRC-32	Message Integrity Code	CBC-MAC	Secure Hash Algorithm
Key Management	Not provided	4-way handshaking mechanism	4-way handshaking mechanism	Simultaneous Authentication of Equals handshark
Authentication	WPE-Open WPE-Shared	Pre-Shared Key(PSK)& 802.1x with EAP variant	Pre-Shared Key(PSK)& 802.1x with EAP variant	Simultaneous Authentication of Equals(SAE)&802.1x with EAP variant

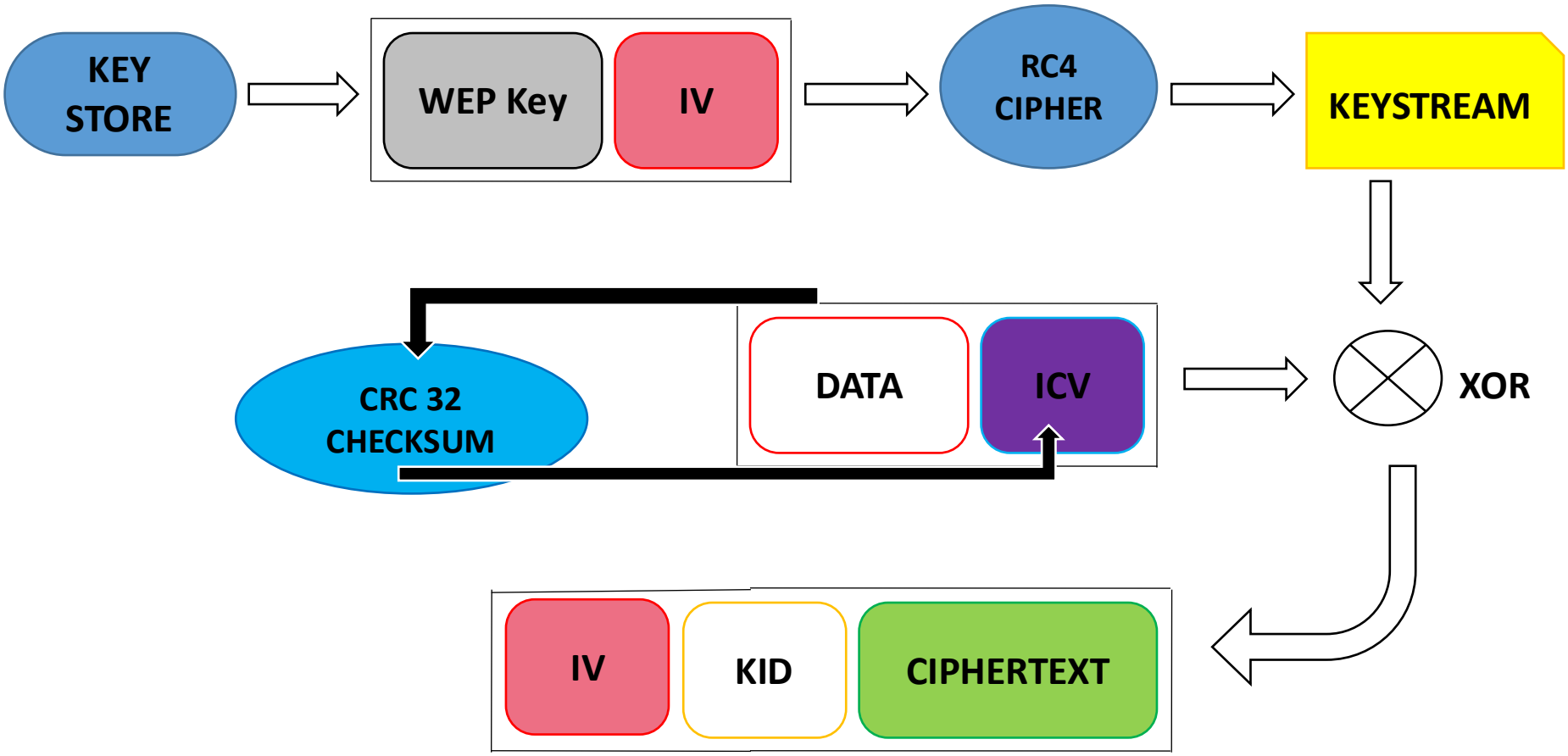
- WEP – Wired Equivalent Privacy encryption standard
- WPA – WiFi Protected Access provided interim fix to WEP vulnerabilities (subset of 802.11i security standard)
- WPA2 (IEEE 802.11i) – official IEEE standard, provided other encryption and authentication mechanisms
- WPA3- latest and safest protocol

Wired Equivalent Privacy (WEP)

1. WEP is based on the RC4 cipher
2. A secret key of 40-bits or 104-bits is combined with a 24-bit Initialisation Vector (IV) to encrypt a plaintext M (Data)
3. The Integrity Check Value (ICV) is computed as the checksum for M
4. The ciphertext, C, is computed as follows:

$$C = [M + ICV(M)] \otimes [RC4(WEP\ Key + IV)]$$

WEP in Operation



WEP ENCRYPTED PACKET

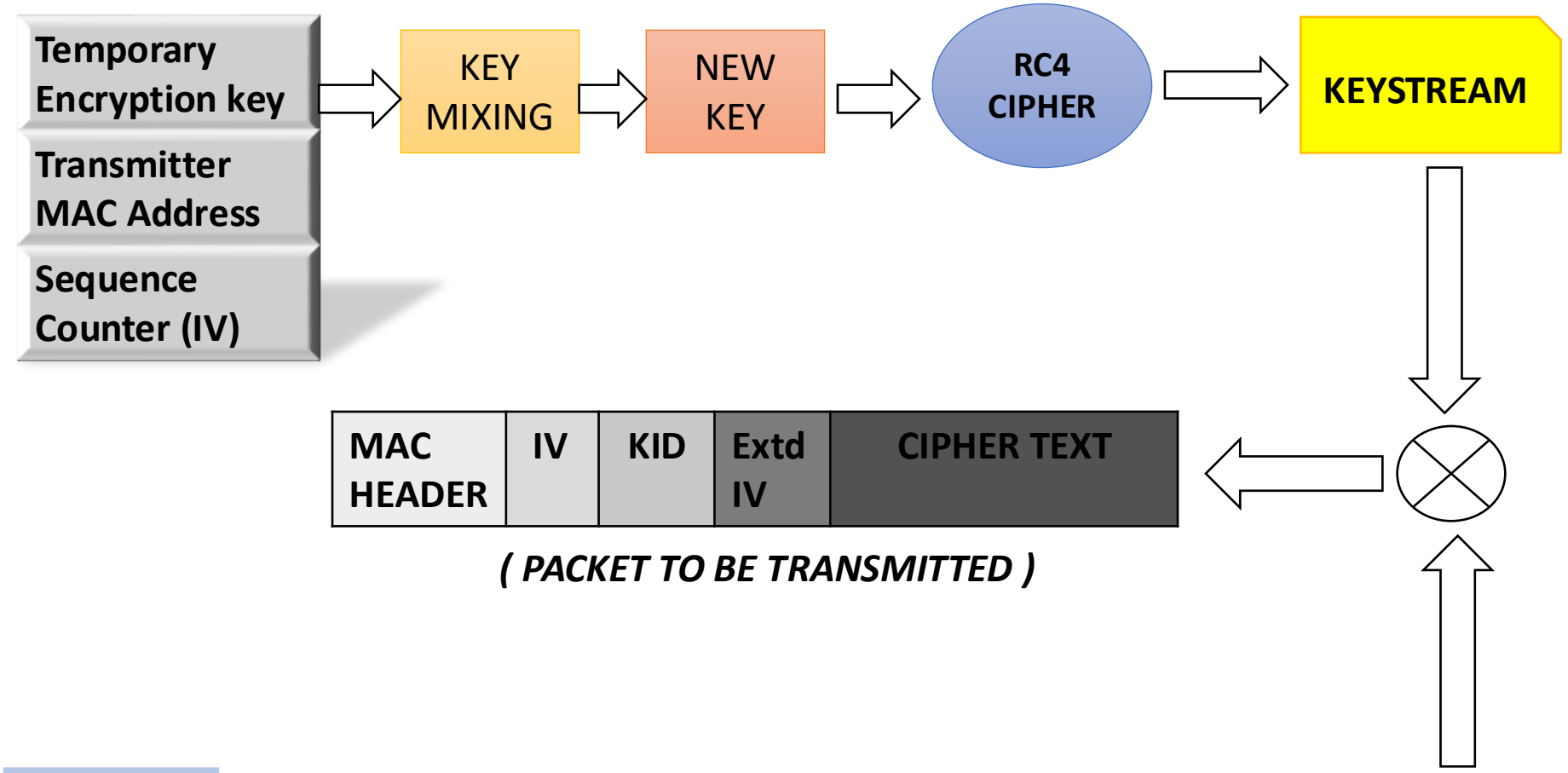
WEP Limitations

1. Key management and key size (WEP key typically 40- or 104-bits) – typically 1 to 4 shared keys used and not changed often
2. Dictionary attack against WEP key generator can be successful within a short time
3. 24 bit Initialization Vector (IV) size – weak security; too short for cryptographic purposes
4. IV reuse produce identical keystreams (different client wireless adapters from same vendor may generate same IV)
5. IV is part of the RC4 encryption key (seeding of RC4 algorithm is weak - details not needed in this module)
6. 32-bit Integrity Check Value (ICV) was used to protect integrity of payload or data in frame but not header

Wi-Fi Protected Access (WPA)

- Subset of the 802.11i standard; Offers improved Encryption & Authentication (e.g. IEEE 802.1X Authentication Server)
- Still based on RC4 Cipher but with 128-bit keys and 48-bit IV
- 48-bit IV may be split into 24-bit IV and 24-bit Extended IV
- **Temporal Key Integrity Protocol (TKIP)** – key different for each user and dynamically changes keys during session (key changes for every frame); every key associated with a sequence number
- The key mixing function takes this sequence number along with the base WPA key and the transmitter MAC address as inputs, and outputs a new key per packet
- This new WPA key is then used along with the IV to generate the keystream; IV also provides key sequence number
- 64-bits Message Integrity Check (MIC) computed by Michael algo; MIC is placed before ICV; with frame counter against Replay attacks

WPA in Operation



WPA Authentication Features

- **WPA Authentication**

- **Pre-shared key (PSK)**
  - every user given the same pass-phrase
  - Uses TKIP less secure than AES
  - preferred for Personal mode - homes, small offices

- **IEEE 802.1X authentication**
  - server distributes different keys to each user
  - enhanced security and authentication
  - preferred for Enterprise mode - business, government, education

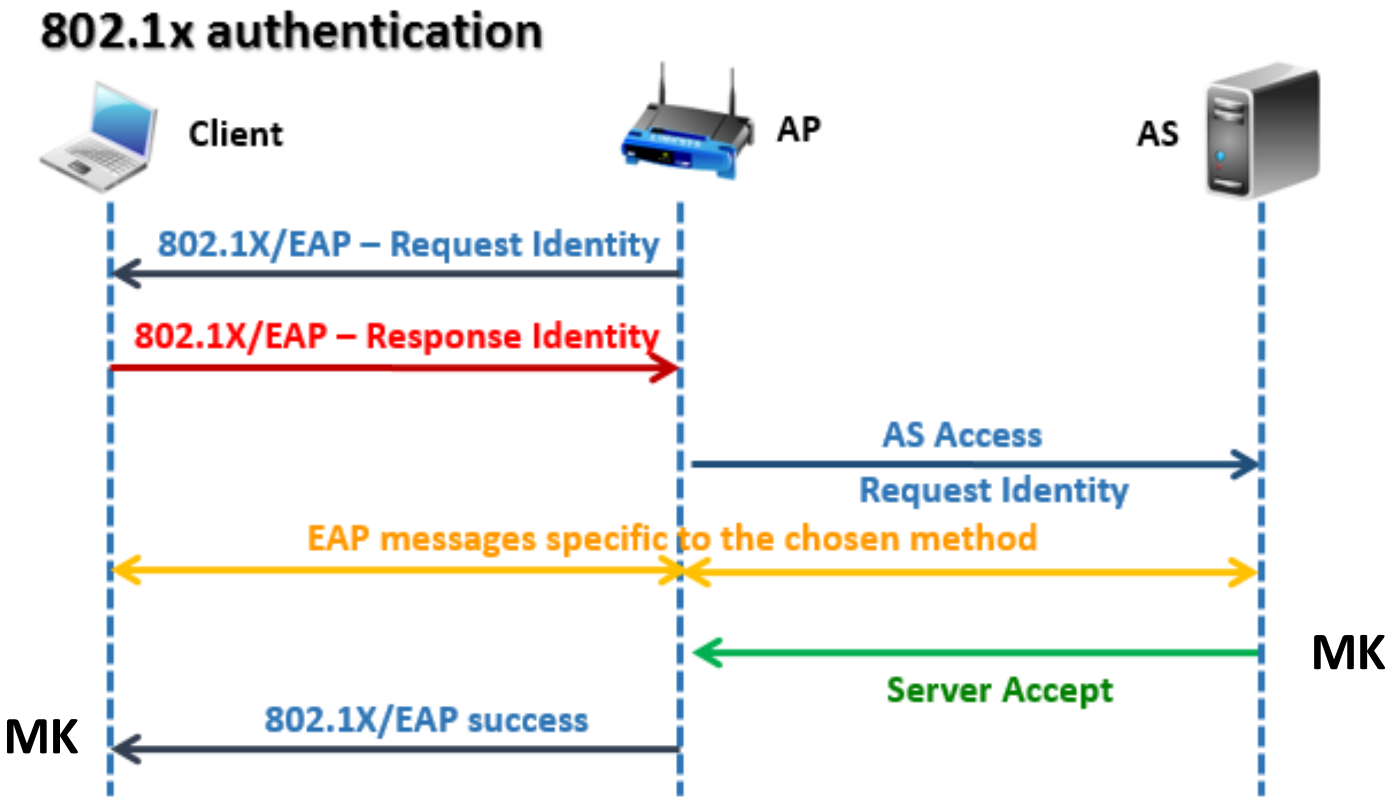
WPA Limitations

- Weaker Authentication in the form of Pre-Shared Key (PSK) – every user given same pass phrase; does not require authentication server
- No standardized way of changing shared secret
- PSK is vulnerable to eavesdropping and dictionary attack
- WPA subject to DoS attacks – flood of EAPOL messages generated requesting 802.1X authentication (authentication server cannot respond)
- Temporal Key Integrity Protocol (TKIP) was a stopgap solution – still uses RC4;
- TKIP vulnerable to keystream recovery attack (Beck-Tews attack) but can only decrypt small portions of data, compromising confidentiality (single ARP frame takes about 14 - 17 minutes to decrypt)

Wi-Fi Protected Access 2 (WPA2)

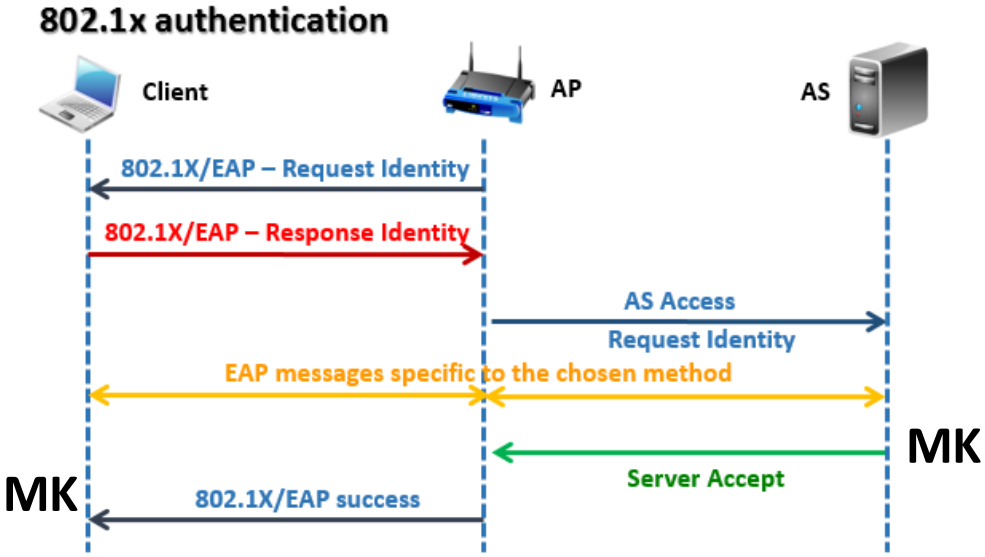
- Uses AES-CCMP (Advanced Encryption Standard – Counter Mode Cipher Block Chaining Message Authentication Code Protocol)
- Authentication similar to WPA (Personal and Enterprise) but using AES-based encryption
- Provides RSN (Robust Security Network)

Robust Security Network via 802.1X



- Protocol to ensure legitimate clients can use WPA2 network
- Three players for 802.1X protocol which uses Extensible Authentication Protocol (EAP) – Client, Access Point(AP) and Authentication Server (AS)
- MK = Master Key

Robust Security Network via 802.1X

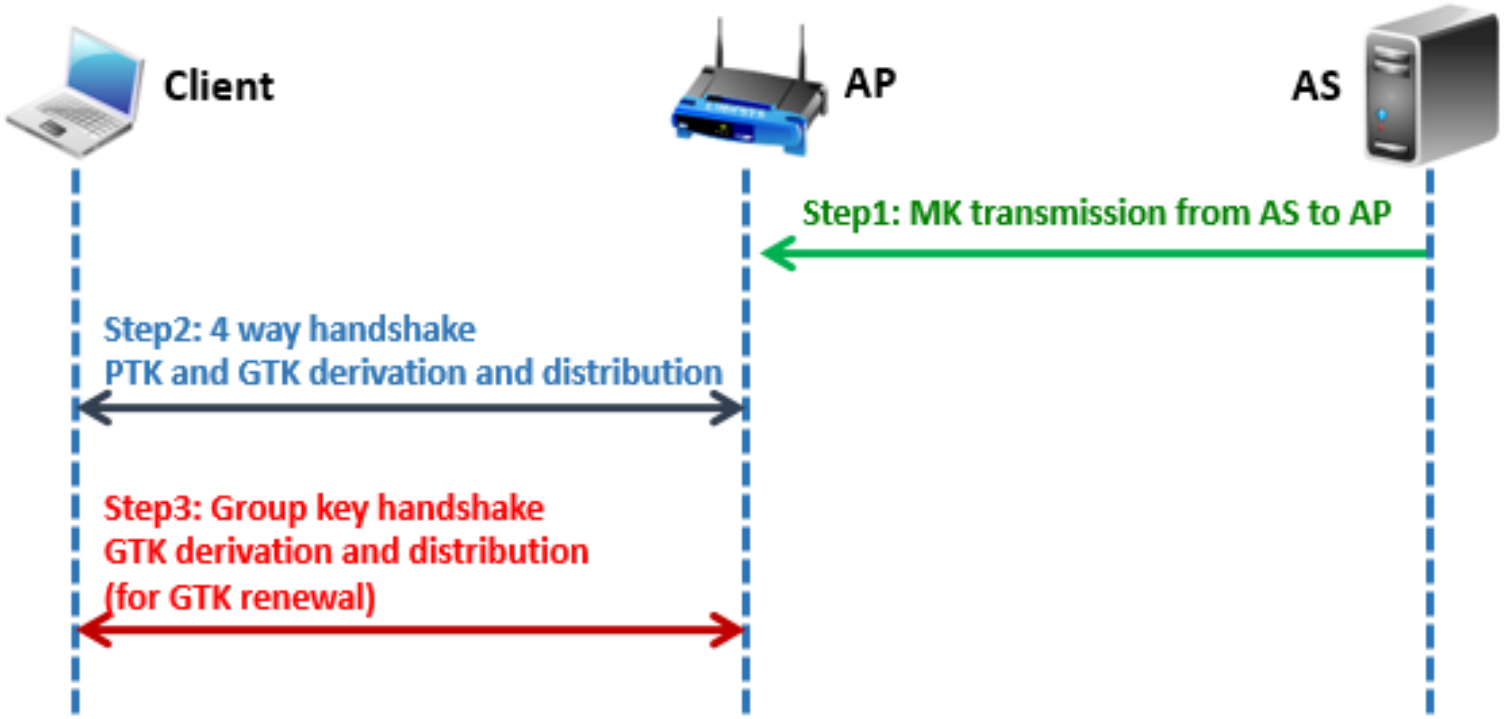


- AP (Access Point) will send "request identity" message to the client, then client will respond with a message containing its identity and preferred authentication method
- AP relays client identity and authentication preferences to AS (Authentication Server)
- After that, EAP messages will be exchanged between the client and the authentication server to generate a common master key (MK).
- At the end of this phase, the authentication server will send a "Server Accept" message to the AP, containing the MK
- The AP then sends a final EAP Success message to the client

WPA2 Key Generation-Distribution

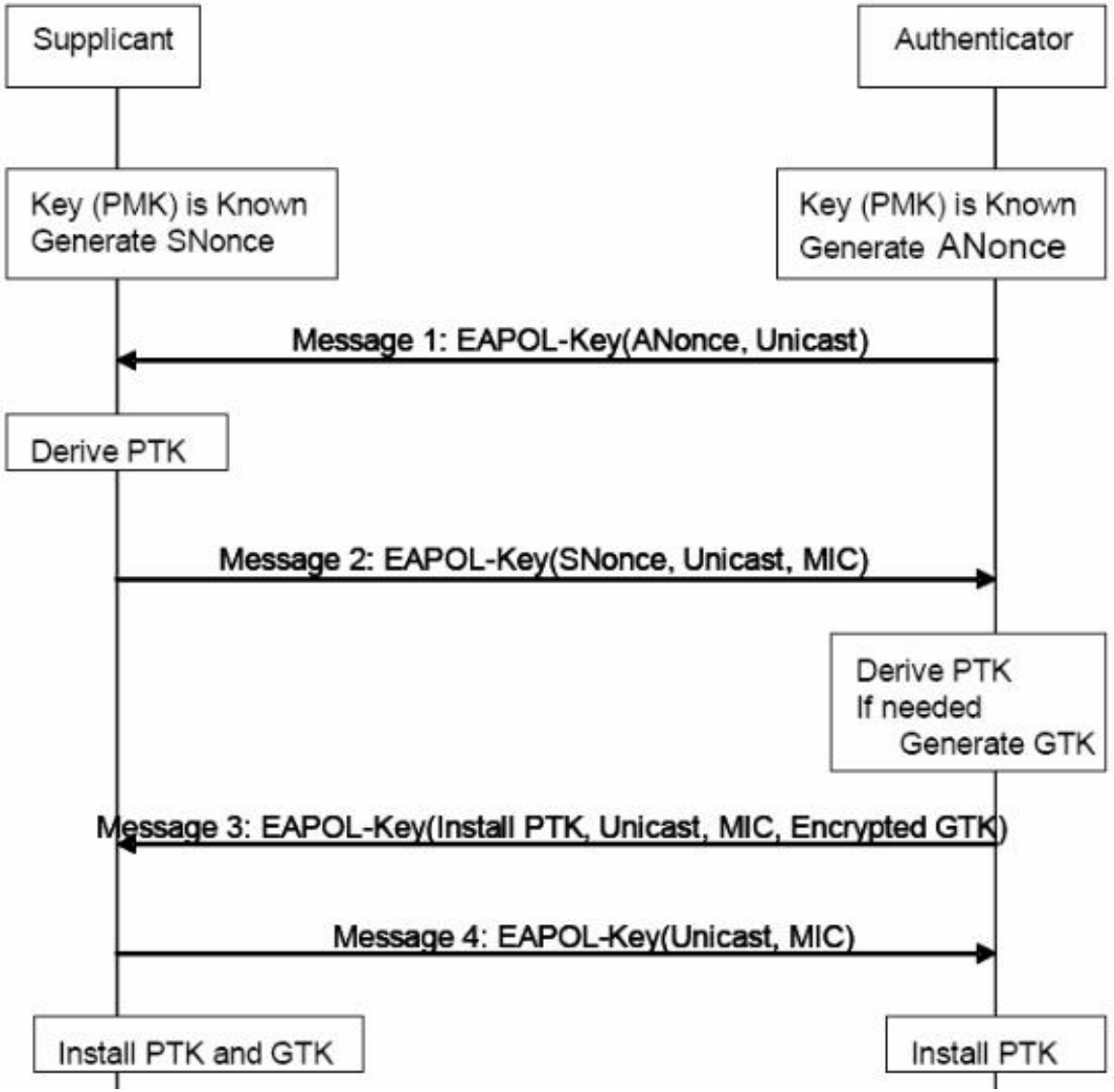
- Authentication similar to WPA (Personal and Enterprise) but using AES instead of RC4
- WPA2-Personal generates a 256-bit pre-shared key from a plain-text pass phrase
- The PSK, SSID and SSID length are used to derive the PMK (pairwise master key) that's used to generate the PTK (pairwise transient key); PTK serves as session key between the wireless client and AP
- WPA2-Enterprise requires an external server called a Remote Authentication Dial In User Service (RADIUS) or AS
- Master key is generated by message exchange between client, AP and AS; client and AP then exchange messages to create the PTK
- The PTK is then used to encrypt and decrypt unicast messages between client and AP

WPA2 Key Generation-Distribution



- AS supplies a different master key to each client
- In both WPA2- Personal and Enterprise, a Group Temporal Key (GTK) is created during the exchange between the client and AP
- GTK then used for encryption-decryption of multicast/broadcast messages

PTK/GTK Generation-Distribution



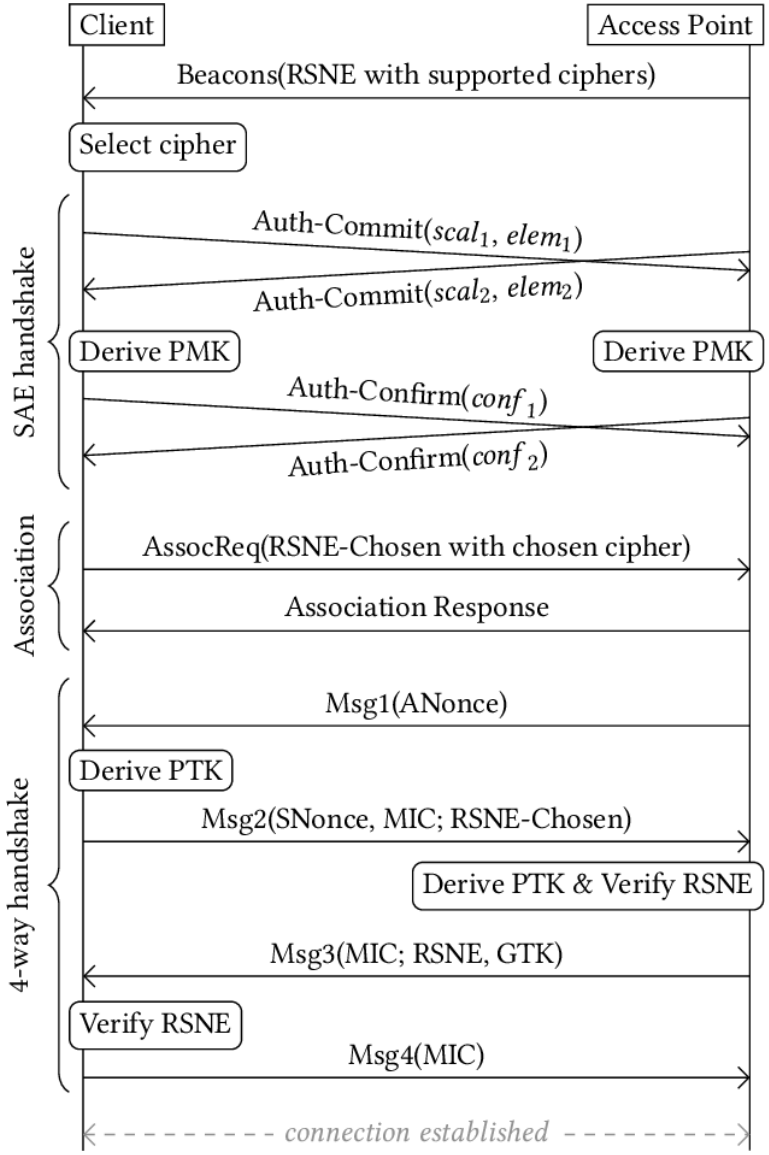
WPA2 Limitations

- Vulnerable to physical layer attacks – e.g. RF Jamming, Data Flooding, Access Point Failure
- Vulnerable to DoS attacks
- Vulnerable to layer 2 session hijacking
- Unprotected control and management frames
- Vulnerable to AP spoofing attacks (but attacker needs to be authenticated client in Enterprise mode)

Wi-Fi Protected Access 3 (WPA3)

- **Secure handshakes**. In old WPA protocols, device and router are connected via a pre-shared key. Unfortunately, this authentication method is relatively easy for hackers to break via brute force attacks. In WPA3 encryption, devices use Simultaneous Authentication of Equals (SAE). It's much harder to eavesdrop passwords, and that could keep devices safer.
- **Stronger protections**. A user can make one offline password guess. Brute-force attacks, in which a hacker uses hundreds of tries, aren't possible.

SAE Protocol



Summary

- Wireless LANs
  - Components – AP, Adapter, Router etc.
  - SSIDs – default values (inherent weakness)
  - Rogue AP – MITM attacks
  - Probes and Beacons – exposes SSIDs and MACs
  - MAC spoofing – bypass address filtering and honeypots
- Security Standards
  - WEP Operation and Limitations
  - WPA Operation, Enhancements and Limitations
  - WPA2 Operation, Enhancements and Limitations

- **Next Lecture – Wireless Attacks, CIA Attack Tools, Defense**

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Wireless Attacks

WIFI Attack Techniques

Attack Type	Description	Techniques/Tools
Rogue AP	Installing an unauthorised AP, creating potential access point into trusted network	Virtual Wifi Router, My Public Wifi, My WIFI Router, Technitium MAC Address Changer
Wardriving	Discovering WLANs by listening to beacons or sending probes, providing launch point for further attacks	Airmon-ng, DStumbler, KisMAC, MacStumbler, NetStumbler, Vistumbler, Wellenreiter, WiFiFoFum
DoS/DDoS	Deny service to a target using one or more attacking stations	HOIC, LOIC, XOIC, HULK
Key Cracking	Capturing data to recover a WPA/WPA2 key	Aircrack-ng, airoway, AirSnort, Wireshark

Understanding Wardriving



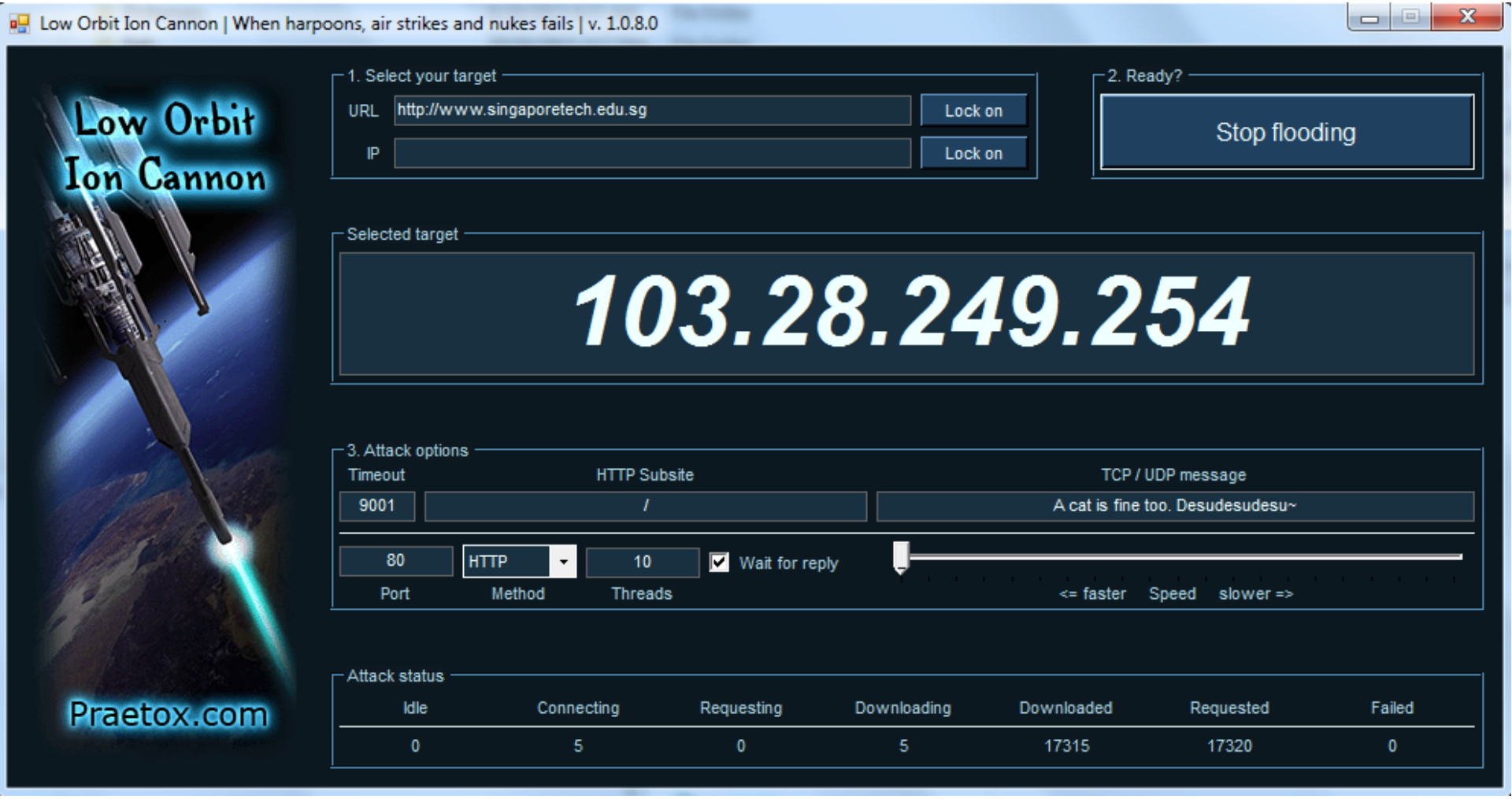


- Detect APs that have not been properly secured
- Many APs have no passwords or security measures, so wardriving can be quite rewarding for attackers
- Wardriving isn't illegal; exploiting discovered network resources without permission is.
- To conduct wardriving, simply drive or move around with a laptop containing a WNIC, an antenna, and software that scans the area for other APs (SSIDs)
- Wardriving has now been expanded to include ***warflying***, which is done by using a drone with an antenna and the same software used in wardriving.

Denial of Service (DoS) Attacks

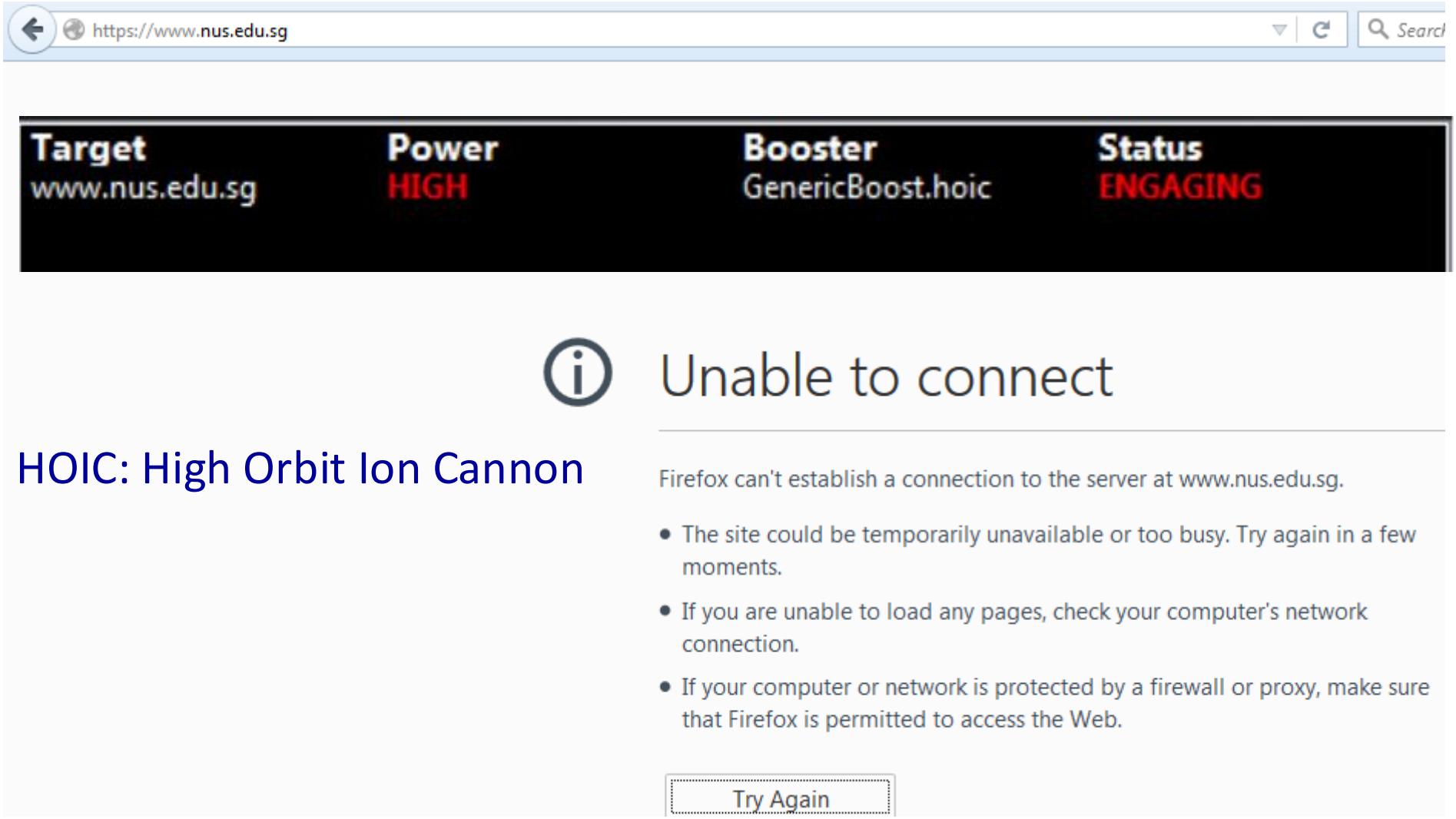
- A DoS attack is an attempt to to deny access to one or more resources, which usually means cutting off access to the service or network as a whole
- This is typically achieved by flooding the system's or server's request queue with fake requests. After this, server will not be able to handle the requests of legitimate users
- In a Distributed DoS (DDoS) attack, many computers start performing DoS attacks on the same target system or server. Attackers typically use a zombie network, which is a group of infected computers on which DOS attacking tool has been installed.
- DOS Attack Tools: LOIC, HOIC, XOIC, HULK, Golden Eye

LOIC DoS/DDoS Demo



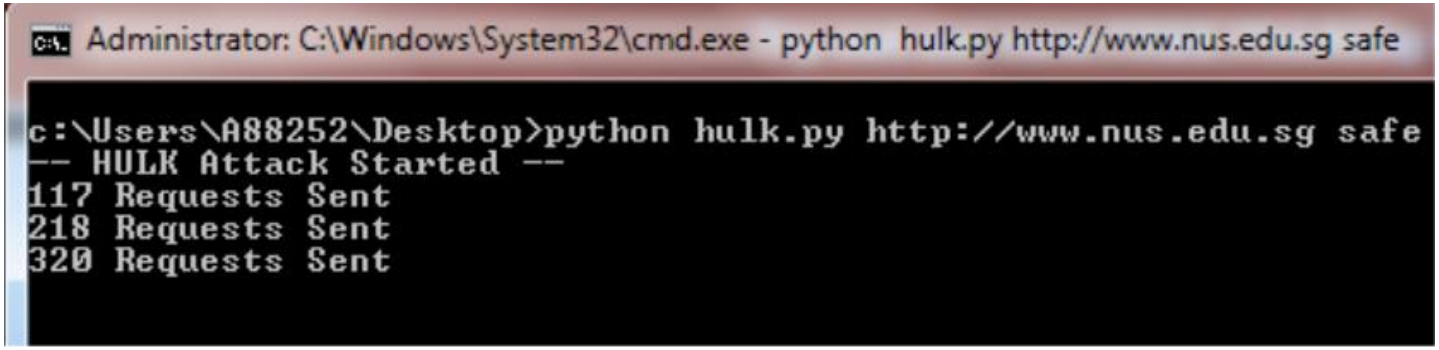
- LOIC: Low Orbit Ion Cannon
- Download LOIC from: http://sourceforge.net/projects/loic/

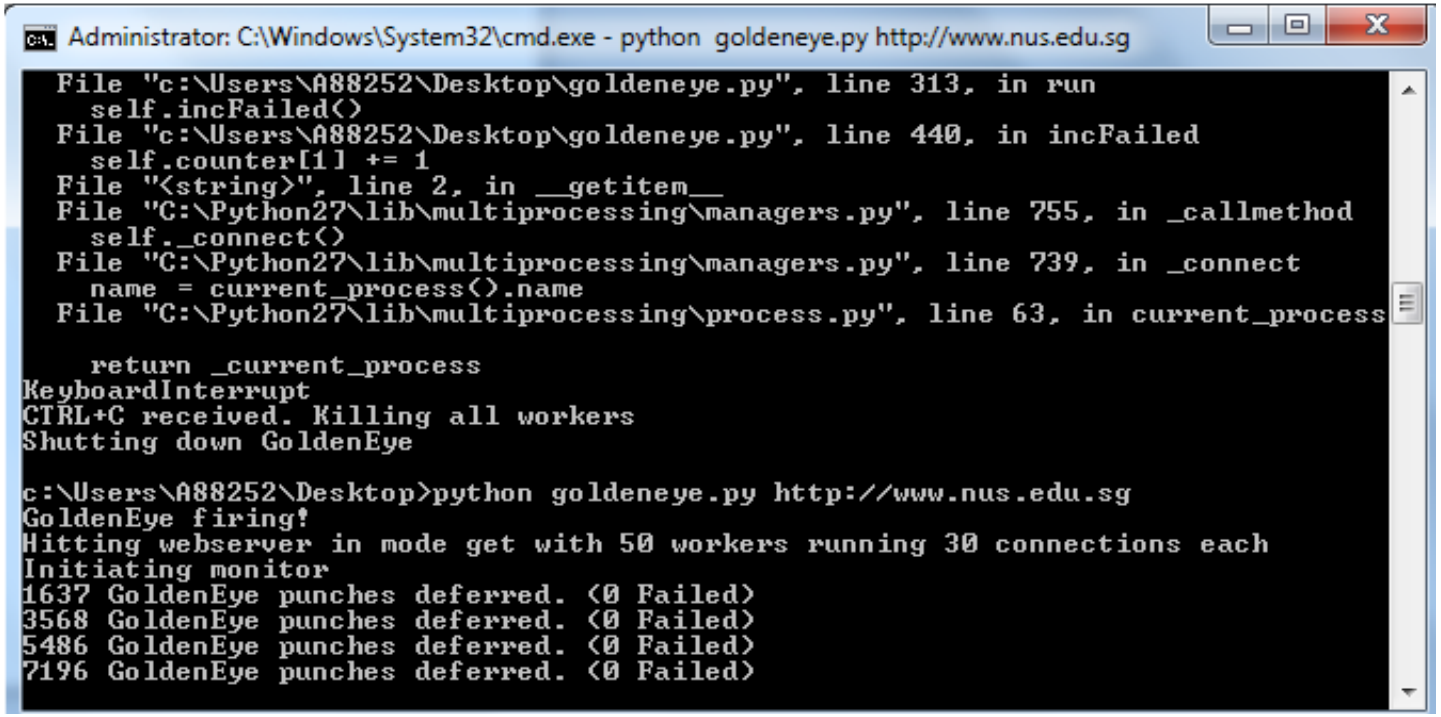
HOIC / Hulk DDoS Demo



HOIC: High Orbit Ion Cannon

Hulk and Golden Eye DDoS Demo





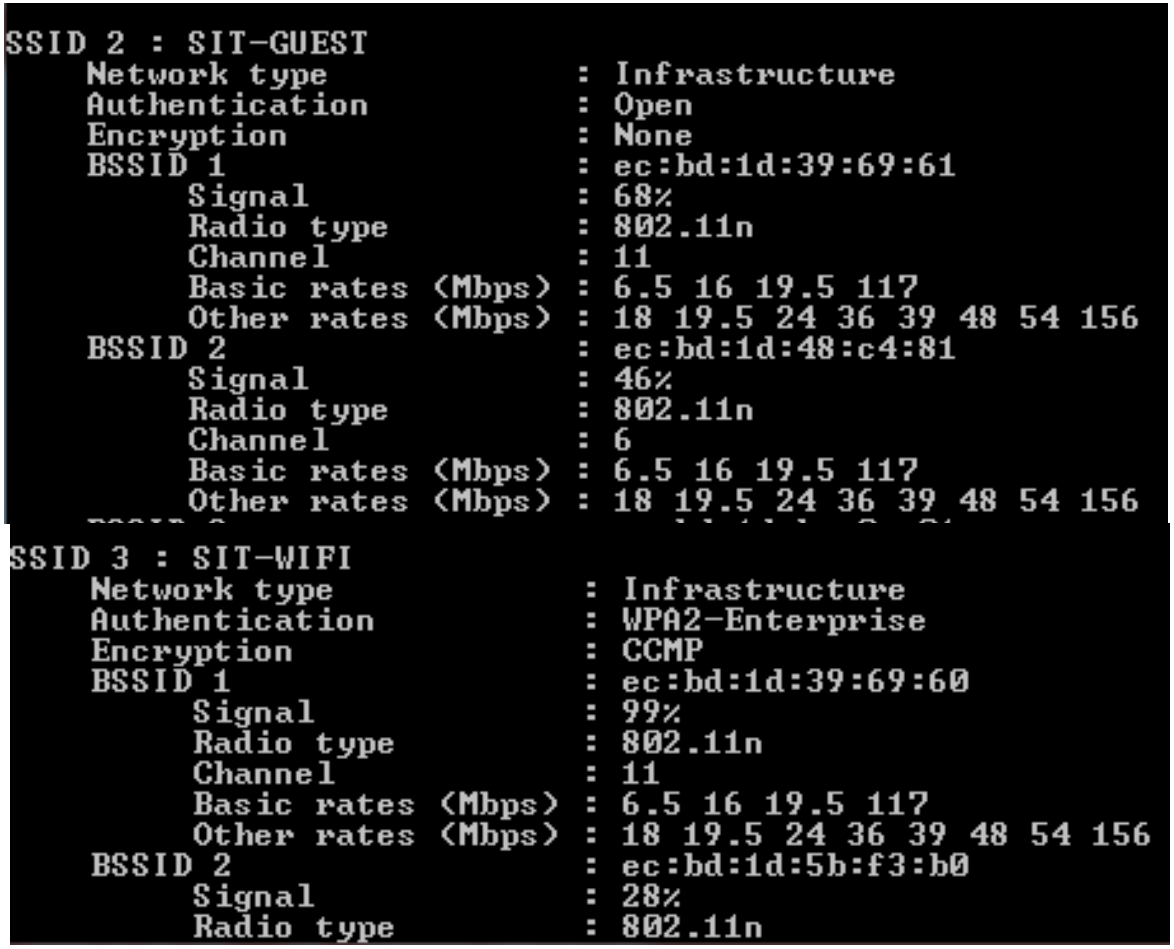
Jamming Attacks

- Jamming occurs when a very powerful radio frequency (RF) signal cuts off the signals from APs and clients in a wireless LAN
- Jamming can be deliberate or unintentional, e.g. when signals from other legitimate devices (in the same frequency range) interfere with WiFi transmissions
- Jamming attacks are similar to DoS attacks
- jamming equipment is expensive so these attacks are rare
- Users need spectrum analyser to find the jamming signal
- My WIFIZone Internet Access Blocker 5.0

Wireless Network Sniffing

- An attacker can passively scan without transmitting at all
- A *passive* scanner instructs the wireless card to listen to specific or all channels – prelude to the remaining attack types
- A WNIC may be configured in RF monitor and promiscuous modes
- A WNIC in monitor mode can capture packets without associating with an AP or ad-hoc network
- Many wireless cards permit RF monitor mode

Wireless Sniffing Demo

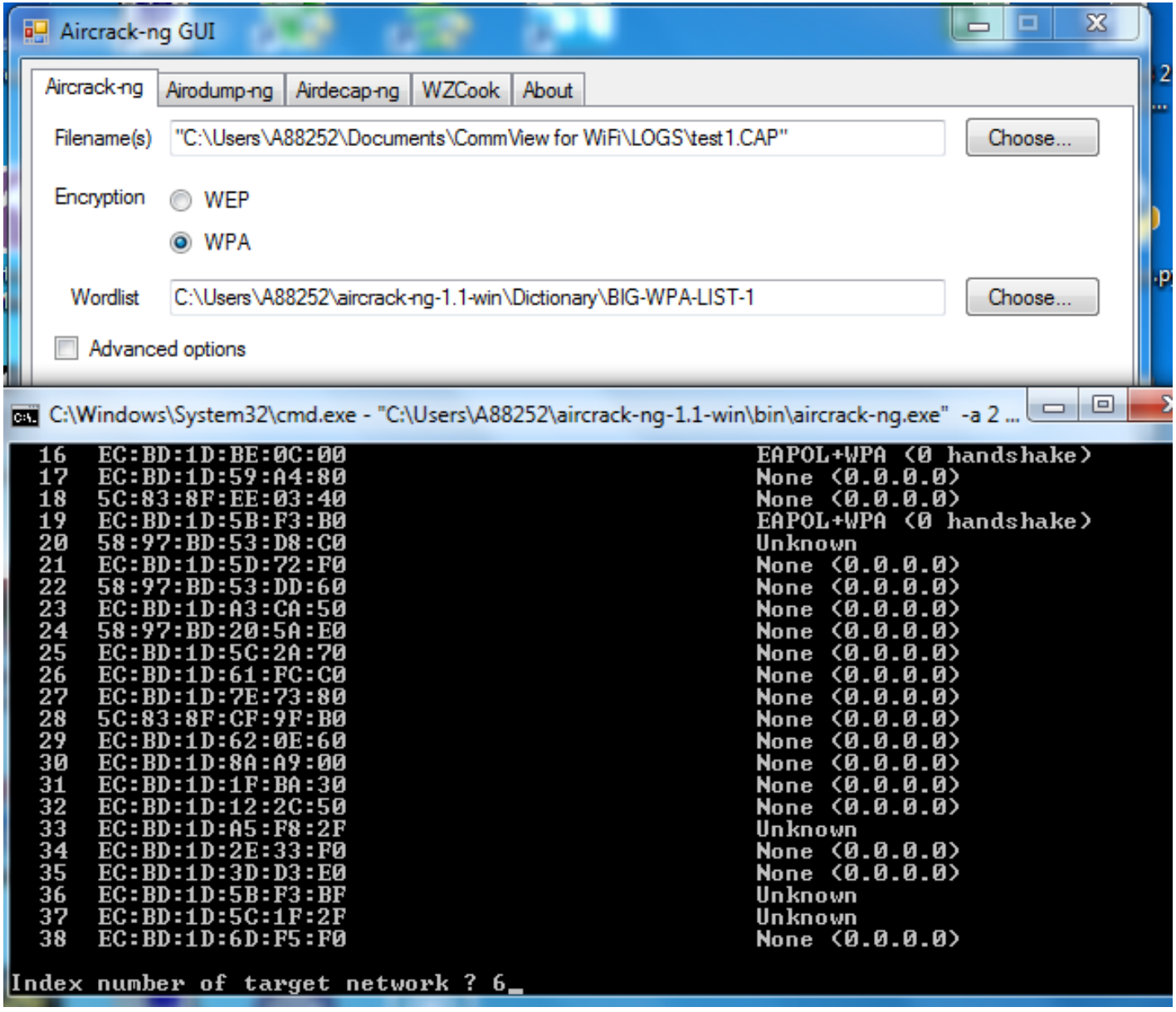


- netsh wlan show networks mode=bssid (scan WiFi neighbourhood)
- BSSID (Basic SSID) is the MAC address of the access point

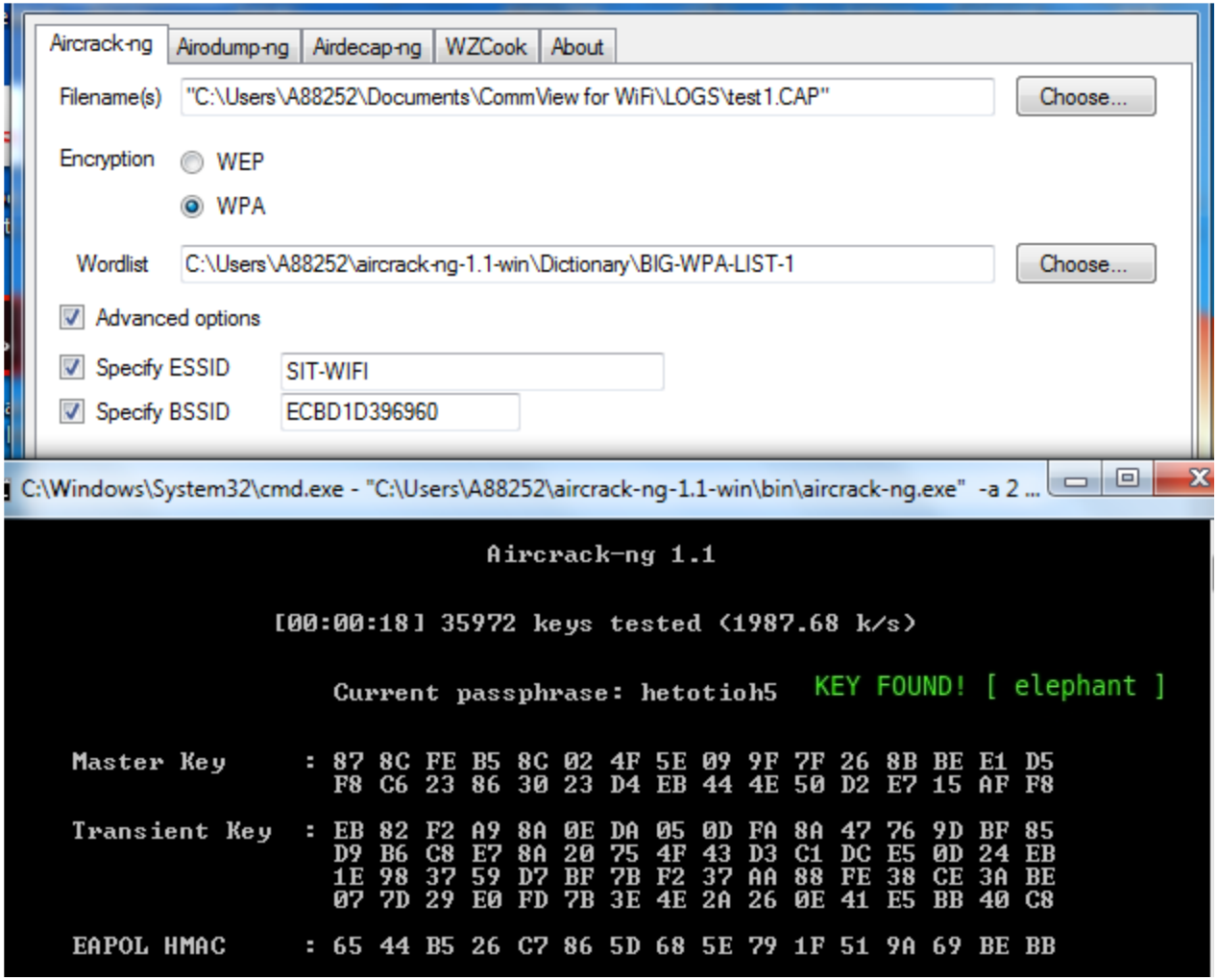
WPA2 Crack via CommView Demo

- Download CommView 7.1 from https://commview.en.softonic.com/download (demo – 5 min scan time)
- Download Aircrack-NG: http://www.aircrack-ng.org/
- Download dictionary: http://download.g0tmi1k.com/wordlists/large/
- Start CommView; set Capture Data Packets only; select Logging tab;
- Logging options – auto-saving; max dir 200000; log file size 100
- Select Nodes tab -> Scanner mode -> all channels -> 0.5s per channel -> Scan
- Select network you want to crack (WPA/WPA2) and change to Single channel mode for selected network's channel
- Wait and try to capture handshake of new device that pops up in the scan
- Stop the scan -> File -> Log Viewer -> File -> Load CommView Logs
- File -> Export Logs -> Wireshark/Tcpdump format <filename.CAP>
- Start Aircrack-NG GUI -> load <filename.CAP> -> select WPA -> load dictionary

WPA2 Crack via CommView Demo



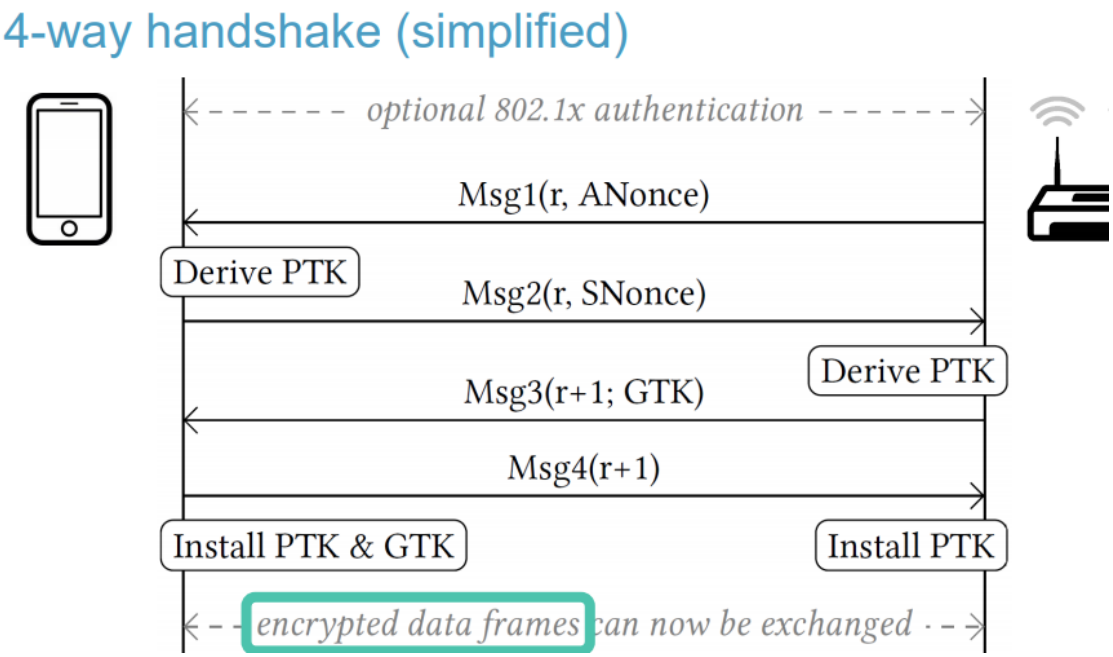
WPA2 Crack via CommView Demo



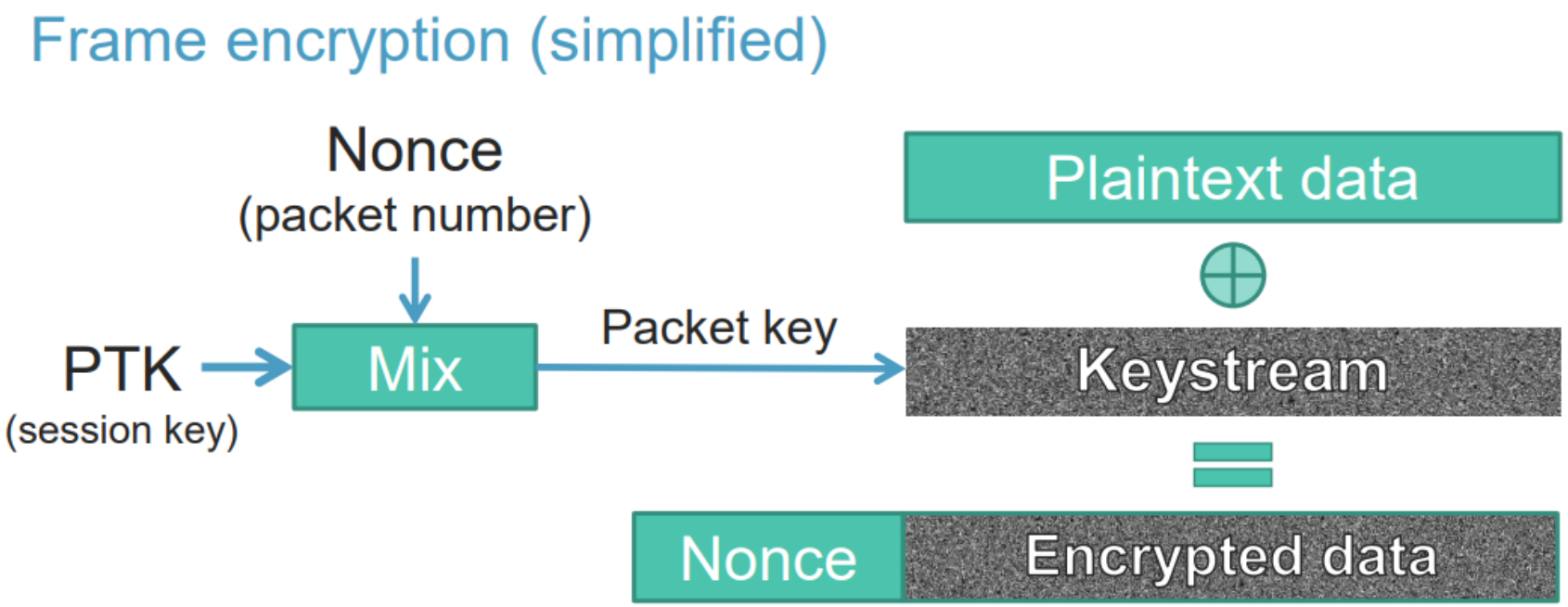
- SSID and BSSID needs too be specified
- Network with handshake needs to be selected
- Use password hesthest

WPA2 Krack Attack

- KRACK is short for (Key Reinstallation Attack)
- Not using brute force attack
- Interrupts the 4-way handshake process to reset the nonce
- Once nonce is reset, it means same nonce is used for encryption and thus creates scope for replay attack
-

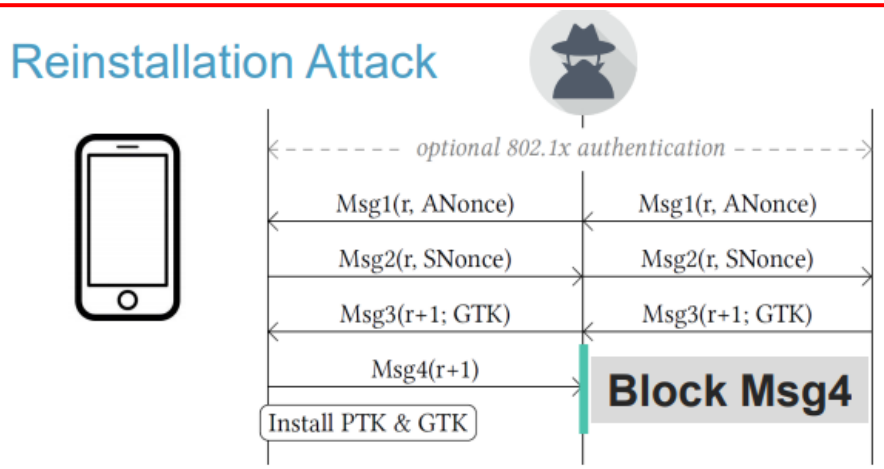


Frame Encryption



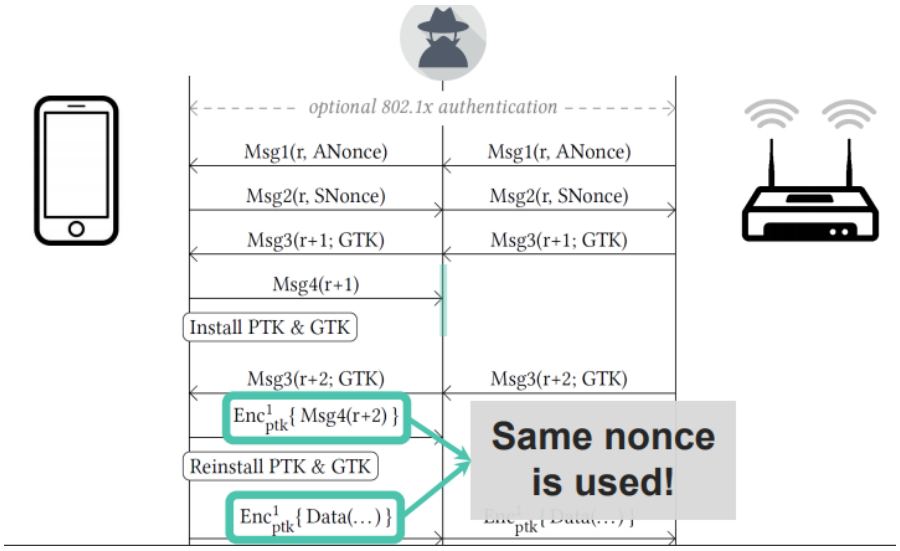
→ Nonce reuse implies keystream reuse (in all WPA2 ciphers)

Key Reinstallation attack



- Block Msg4 from User
- This makes AP resend Msg3

- When user receives Msg3, it reinstalls the key
- Nonce is reset
- On some devices Nonce changes to 0



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CIA Attack Tools

Confidentiality Attacking Tools

Tools	Description	Type of Attack
AirSnort	Brute force WEP cracker	Encryption Cracker
AirCrack	WPA cracker	Encryption Cracker
Ettercap, dsniff, and Wireshark	Packet sniffers with traffic analysis. These also include tools to break encryption.	Packet sniffing
Hotspotter, APsniff, APhunter, and KNSGEM	Discovers WLANs by listening for beacon signals transmitted from APs.	AP locator
HermesAP and OpenAP	Used to setup an rogue AP	Evil Twin
OpenWRT and HyperWRT	Replacement firmware so APs can be programmed to execute attacks.	Fake AP creation

Integrity Attacking Tools

Tools	Description	Type of Attack
Airpwn	Allows for generic 802.11 packet injection	802.11 packet injection
File2air	Allow the specified file be used as packet payload.	802.11 replay
AirJack and Simple-replay	Allows previously captured packets to be injected back into the network.	802.11 replay

Availability Attacking Tools

Tools	Description	Type of Attack
FakeAP	Generate thousands of 802.11 beacon signals.	Flooding DoS
Void11	Can be used to execute deauthenticate, authenticate, and association flooding attack.	Flooding DoS
Many commercial tools available	Jams the RF signal so that it cannot be distinguished by a legitimate device.	RF jamming

Bluetooth Attacking Tools

Tools	Description	Type of Attack
BlueSmack	Issues ping of death attack	DoS
BlueChop	Disrupts and existing piconet	DoS
BluePass	Causes a buffer overflow attack	DoS
BlueSnarf	Obtain unauthorized access to files.	Bluesnarfing
BlueBump	Obtains the piconet key	Key bump

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Defense

WIFI Security Measures

- Change the default router authentication
- If your equipment supports it, use WPA or WPA2 because it offers better encryption than WEP
- Always check for updates to your router
- Use a router to filter unauthorized MAC addresses and prevent them from having network access (can spoof)
- Change default AP settings e.g. default SSIDs and disable broadcast
- Measures to prevent radio waves from leaving or entering a building e.g. WiFi-proof paint, WiFi-proof windows, Faraday cages, etc.
- Turn off your router or AP when not using it

Summary

- Wireless Attacking Techniques
  - Rogue AP (MAC Spoofing), Wardriving
    DoS/DDoS, WPA2 Cracking (need computation power)
- CIA Tools
  - Public domain tools that can launch attacks on Confidentiality, Integrity as well as Availability of the target
- Countermeasures
  - WIFI Security Best Practices
  - Available tool support

- **Next Lecture – Web Hacking**